

Weighted Regularization in Fixed Effects Panel Model: Convergence Rate and Oracle Property

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Abstract

In this paper, we study a weighted ℓ_1 -penalized fixed effects estimator for the classical linear panel data model and investigate its theoretical properties. The proposed estimator minimizes the sum of squared residuals plus an ℓ_1 -penalty on the fixed effects, weighted by a preliminary estimate of these effects. We derive a non-asymptotic ℓ_2 -error bound for the estimator and show that it is asymptotically normal, with the same mean and variance as if the true model were known in advance, under suitable restrictions on the growth rate of the number of significant parameters. Our study extends the results of [Huang et al. \(2008b\)](#) to a panel data setting, albeit under some restrictive assumptions. To complement the theoretical findings, we provide simulation evidence and revisit the empirical analysis of [Donohue III and Levitt \(2001\)](#) to illustrate a real-world application.

1 Introduction

In this paper, we consider a weighted regularized estimator for the following linear one-way error components model:

$$y_{it} = \mathbf{x}_{it}'\boldsymbol{\beta}^* + \mathbf{d}_i'\boldsymbol{\alpha}^* + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad (1)$$

where $\mathbf{x}_{it}$ is a $p \times 1$ vector of covariates with fixed dimension, and $\boldsymbol{\beta}^*$ denotes the true population parameter of interest. The vector $\mathbf{d}_i$ is an $N \times 1$ individual indicator vector, that takes the value 1 at the i th coordinate and zero elsewhere, and $\boldsymbol{\alpha}^*$ is an $N \times 1$ vector of individual-specific parameters. The individual time-invariant heterogeneity (e.g., intelligence, ability, or motivation) is captured through the term $\mathbf{d}_i'\boldsymbol{\alpha}^* = \alpha_i$, while the ε_{it} are the idiosyncratic errors, which we shall be more specific later.

Two standard estimators for $\boldsymbol{\beta}^*$ are the random effects estimator and the fixed effects estimator ([Hsiao, 2022](#)). The random effects estimator is efficient under specific assumptions regarding the stochastic structure of $\boldsymbol{\alpha}^*$. However, as we demonstrate in our simulation studies, the random effects estimator can be extremely biased even under minor violations of the condition $\mathbb{E}[\alpha_i^* \mathbf{x}_{it}] = 0$. Current econometric practice tends to favor robustness over efficiency; consequently, a more popular approach for estimating $\boldsymbol{\beta}^*$ is the fixed effects estimator, which treat the individual heterogeneity as 'fixed' regressors, thereby avoiding assumptions about their stochastic relationship

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with the other variables. We denote the fixed effects estimator as follows:

$$\hat{\beta}^{fe} := (\mathbf{X}'\mathbf{R}_D\mathbf{X})^{-1}\mathbf{X}'\mathbf{R}_D\mathbf{y}, \quad (2)$$

where the $\mathbf{X}$ and $\mathbf{y}$ are obtained by stacking $\mathbf{x}'_{it}$ and y_{it} , respectively. The matrix $\mathbf{D}$ denotes the individual indicator matrix, and $\mathbf{R}_D = \mathbf{I}_{NT} - \mathbf{D}(\mathbf{D}'\mathbf{D})^{-1}\mathbf{D}'$ is the projection matrix that removes individual fixed effects. The fixed effects estimator is robust to arbitrary dependence between $\mathbf{x}_{it}$ and any time-invariant omitted variables, as it controls for such heterogeneity directly in the regression. This identification strategy offers a significant advantage over alternative methods. The transformation $\mathbf{R}_D$ is then applied to eliminate the fixed effects from the model and thereby avoid the incidental parameters problem associated with the diverging dimension of α^* .

To motivate the proposed estimator in this paper, we make two simple observations of the $\hat{\beta}^{fe}$. First, note that $\mathbf{X}'\mathbf{X} \geq \mathbf{X}'\mathbf{R}_D\mathbf{X}$ by the property of the projection matrix,¹ which implies that the transformation $\mathbf{R}_D$ reduces the efficiency of the estimator, since the inverse of the latter matrix appears in the formula for the variance of the fixed effects estimator. Another way to see this is that the transformation $\mathbf{R}_D$ removes the individual averages from the regressors $\mathbf{x}_{it}$, and therefore the estimation variance increases due to the reduced variation in the regressors. This fact will be evident in our simulation studies. Secondly, in most applications, researchers are only interested in the coefficients β^* , and one can compute $\hat{\beta}$ without worrying about α^* . A simple estimator for the individual fixed effects can be calculated using the formula

$$\hat{\alpha}_i^{fe} := \frac{1}{T} \left(\sum_{t=1}^T y_{it} - \sum_{t=1}^T \mathbf{x}'_{it} \hat{\beta}^{fe} \right). \quad (3)$$

Exploiting the information contained in $\hat{\alpha}^{fe}$ could potentially lead to a more efficient estimating procedure.

In this paper, we explore the use of a classical idea from the machine learning literature to improve the performance of the fixed effects estimator. We define the **Sparse Fixed Effects Estimator**, denoted by $\hat{\beta}^{sfe}$ and $\hat{\alpha}^{sfe}$, as the minimizer of the sum of squared errors plus a penalty term on the size of the individual fixed effects, weighted by a first-stage estimator. Similar to the canonical fixed effects estimator, individual heterogeneity is controlled directly by including an individual indicator vector in the regression. As a result, the estimator inherits robustness to dependence between the covariates and unobserved individual fixed effects. The ℓ_1 -regularization is introduced to handle the diverging dimensionality of the parameters and to reduce the high variance caused by estimating a high-dimensional parameter. Finally, we use $\hat{\alpha}^{fe}$ to construct the weights to correct for the bias induced by the shrinkage effect of the ℓ_1 penalty, which is a common drawback of regularized estimators. We emphasize that these weights are crucial for estimating β^* when penalizing the fixed effects, as they substantially reduce shrinkage bias and improve the model's selection consistency in finite sample simulation studies. Theoretically, the use of weights also enables the sparse fixed effects estimator to achieve favorable asymptotic properties, which are generally unattainable under unweighted regularization. The intuition behind these weights is simple: by defining the weights as a function of a consistent estimator of α^* , one can theoretically discriminate between the penalties imposed on zero and nonzero coefficients. In large samples, the weights become sufficiently accurate such that the nonzero coefficients receive

¹For matrices $\mathbf{A}$ and $\mathbf{B}$, $\mathbf{A} \geq \mathbf{B}$ if and only if $\mathbf{A} - \mathbf{B}$ is positive semi-definite.

a bounded penalty, while the zero coefficients receive an unbounded penalty. As a result, the shrinkage bias induced by regularization is minimized.

The proposed estimator can be reformulated as a Lasso-type minimization problem, and its solution can be efficiently obtained using the Coordinate Descent Algorithm, with the same computational cost as the ordinary least squares estimator (Tseng, 2001). Due to its low computational cost, we are able to conduct large-scale simulation studies. In the simulation analysis, we compare the proposed estimator with several common estimation strategies: the fixed effects estimator (OLS after the within transformation), the random effects estimator (generalized least squares), standard Lasso (unweighted regularization), and the post-selection estimator (refitting the model after variable selection). We show that the proposed estimator outperforms these alternatives in many settings, as measured by the empirical risk function. We also compare the proposed estimator to the oracle estimator. The simulation results demonstrate that the proposed estimator closely approximates the performance of the oracle estimator across a wide range of settings.

We contribute to the literature by providing a complete characterization of the theoretical properties of weighted regularization in the context of panel data with a diverging number of parameters. In doing so, we improve the classical estimation procedure by incorporating ideas from the machine learning literature. In particular, (1) Theorem 1 establishes a non-asymptotic rate of convergence for the ℓ_2 -error of the proposed estimator under relatively weak assumptions; and (2) Theorem 3 demonstrates the oracle property: the proposed estimator is asymptotically normal, with the same mean and covariance as if the true population model were known in advance, under suitable conditions on the growth rate of the number of significant parameters, the tuning parameter, and the sample size, along with other regularity conditions.

1.1 Related Literature

Estimators with the ℓ_1 penalty were first studied by Tibshirani (1996). Fu and Knight (2000) investigated the asymptotic properties of estimators with an ℓ_q -penalty, for $0 < q \leq 2$, when the number of the parameters is fixed. They showed that, under appropriate conditions, the estimator with an ℓ_1 -penalty is consistent, and its limiting distribution can have positive probability mass at zero if the true parameter value is zero.

Penalties beyond the ℓ_q -norm have also been proposed. One notable example is the smoothly clipped absolute deviation (SCAD) penalty introduced by Fan and Li (2001), who showed that SCAD estimators possess the oracle property—meaning the estimator behaves asymptotically as if the true underlying model were known in advance. Subsequently, Zou (2006); Meinshausen and Bühlmann (2006); Zhao and Yu (2006) provided evidence around the same period that the standard ℓ_1 -penalized estimator (lasso) does not have the oracle property in general. While Meinshausen and Bühlmann (2006) and Zhao and Yu (2006) outlined conditions under which the lasso estimator is variable-selection consistent, Zou (2006) proposed a weighted ℓ_1 -penalty (adaptive lasso) and demonstrated its oracle property when the number of parameters is fixed.

Huang et al. (2008a) extended the analysis of Fu and Knight (2000) to high-dimensional settings and established the oracle properties of ℓ_q -penalized estimators with $q < 1$. Kock (2013) further generalized these results to panel data contexts under the assumption of a fixed number of parameters, allowing for random covariates.

The oracle property of high-dimensional weighted penalized estimators was also rigorously proven by Huang et al. (2008b). In this paper, we extend the results in Huang et al. (2008b) to

panel data models where we regularize high-dimensional nuisance parameters using weights derived from a preliminary estimator that leverages the panel data structure. Specifically, we derive convergence rates under relatively mild assumptions and establish the oracle property of our proposed estimator under conditions similar to those in [Huang et al. \(2008b\)](#).

The penalized approach for estimating fixed effects in panel models was first studied by [Koenker \(2004\)](#), and later extended to dynamic panel models by [Galvao and Montes-Rojas \(2010\)](#) in the context of quantile regression. These studies established asymptotic normality for the proposed estimators, but with a biased center. Due to the nonlinear nature of the model, it is difficult to disentangle the bias introduced by regularization from other sources of bias. In this paper, we propose an improved regularization strategy for fixed effects estimation—albeit under a simpler setting—that is asymptotically free from shrinkage bias.

The rest of the paper is organized as follows. In Section 2, we formally define the Sparse Fixed Effects estimator and discuss its computation, including the construction of the weights and the selection of the penalty parameter. In Section 3, we present the main theoretical results of the paper: we derive the rate of convergence and establish the asymptotic distribution of the proposed estimator. Section 4 reports the results of simulation studies, where we compare the Sparse Fixed Effects estimator with other commonly used panel data estimators. In Section 5, we revisit the study by [Donohue III and Levitt \(2001\)](#) on the effects of legalized abortion on crime rates. Lastly, Section 6 concludes. Proofs and additional simulation results are provided in the appendix.

2 Sparse Fixed Effects Estimator

In this section, we introduce the Sparse Fixed Effects Estimator for the one-way error components model from a methodological point of view. We consider a panel data structure consisting of observations on N cross-sectional units (e.g., individuals, firms, countries, or states) over T time periods, totaling $n = NT$ observations. Our goal is to estimate the effect of a p -dimensional regressor $\mathbf{x}_{it}$ on a scalar outcome variable y_{it} in the presence of unobserved individual-specific heterogeneity. The relationship is modeled by equation (1), where β^* is a $p \times 1$ population parameter of interest, and α^* is an $N \times 1$ nuisance parameter that must be controlled for consistent estimation of β^* . We note that the sparsity structure of α^* is irrelevant in the estimation.

We formally define the Sparse Fixed Effects (SFE) estimators $\hat{\beta}^{sfe}$ and $\hat{\alpha}^{sfe}$ as²

$$\hat{\beta}, \hat{\alpha} := \underset{\beta \in \mathbb{R}^p, \alpha \in \mathbb{R}^N}{\operatorname{argmin}} \sum_{i=1}^N \sum_{t=1}^T (y_{it} - \mathbf{x}_{it}'\beta - \mathbf{d}_i'\alpha)^2 + \lambda_n \sum_{i=1}^N w_i |\alpha_i|, \quad (4)$$

where $\lambda_n \geq 0$ is a tuning parameter that is selected separately from the above estimation procedure, and $w_i \geq 0$ is an individual-specific, data-dependent weight determined by a first-stage estimator. We also express the estimator in a more compact matrix notation:

$$\hat{\beta}, \hat{\alpha} := \underset{\beta \in \mathbb{R}^p, \alpha \in \mathbb{R}^N}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{X}\beta - \mathbf{D}\alpha\|_2^2 + \lambda_n \|\mathbf{W}\alpha\|_1, \quad (5)$$

where $\mathbf{w} := (w_1, \dots, w_N)'$ and $\mathbf{W} := \operatorname{diag}(\mathbf{w})$. This formulation will be convenient for some of the later discussions. For the remainder of this section, we make a few remarks on the features of the

²We will drop the superscript *sfe* from this point forward for notational simplicity.

proposed estimator.

2.1 Validity of Sparsity

Although the estimator defined in equation (4) can be applied regardless of the structure of the true population parameter α^* , simple intuition suggests that the regularized estimator will perform better when the "size" of α^* is small. In other words, we would ideally hope that the true parameter α^* satisfies some form of sparsity assumption in order to justify the use of regularization. This point will be addressed more rigorously in the following sections.

Whether the true population parameter α^* is "sparse"³ is ultimately an empirical question, and such an assumption is generally difficult to verify. Nevertheless, we offer a few comments on the plausibility of assuming that α^* is sparse in the current context.

In a typical empirical study involving a linear specification, it is common to include a constant intercept term μ , which can be easily estimated by augmenting the covariate matrix $\mathbf{X}$ with a column of ones. However, the intercept term μ is not identified in model (1), since the sum of all columns in $\mathbf{D}$ equals a vector of ones. To resolve this perfect multicollinearity, a common approach is to drop the column of ones from the covariate matrix. Consequently, for each individual i , the model identifies the combined individual-specific intercept $\mu + \alpha_i^*$, rather than α_i^* alone. In terms of estimating β^* , this is equivalent to imposing the prior restriction $\sum_{i=1}^N \alpha_i = 0$ when minimizing equation (4) (see the discussion in Chapter 2.2 of [Hsiao \(2022\)](#)). This restriction offers a convenient interpretation: α_i^* can be viewed as the individual-specific deviation of unit i from the population intercept. Under this interpretation, it is conceptually natural to assume that the distribution of these deviations is centered around zero, particularly in the case where the population intercept $\mu = 0$. This can be achieved by recentering all variables, a common practice in machine learning, though originally intended for different purposes.

To summarize, although the sparsity assumption on α_i^* may not be fully realistic, by treating α_i^* as the deviation from the population intercept and recentering all variables, it is not unreasonable to assume that the distribution of α_i^* is centered around zero. Indeed, this latter assumption is also imposed in the random effects framework.

2.2 Feasible Weights

The idea of using weights to improve the performance of regularized estimators was first introduced by the seminal paper [Zou \(2006\)](#). In the original paper, [Zou \(2006\)](#) proposed using the inverse of the absolute value of the ordinary least squares estimator as the weight in the penalty term when the dimension of the regressors is small, so that the conventional estimator remains computationally feasible. For the high-dimensional case ($p \geq n$), the author suggested using the ridge estimator to construct the weights. However, due to the shrinkage bias of the ridge estimator, there is no theoretical guarantee for its validity. Indeed, [Zou \(2006\)](#) pointed out that the choice of the first-stage estimator can be quite flexible. Later, [Huang et al. \(2008b\)](#) showed that when $p > n$, marginal regression can be used to construct the weights with theoretical justification.

³The notion of sparsity of a vector can be defined rigorously by restricting the size of its ℓ_q -norm, where $q \geq 0$. In the later asymptotic analysis, instead of using this definition, our results work directly with the number of nonzero elements in α^* , denoted by s_N . Also note that when $q < 1$, the " ℓ_q -norm" is only nominally referred to as a norm.

In the current context, a natural and computationally convenient choice of weights exists. Specifically, we define the weights as

$$w_i = \frac{1}{|\hat{\alpha}_i^{fe}|} \quad (6)$$

for $i = 1, \dots, N$. Note that the formula we use to calculate $\hat{\alpha}$ is equivalent to the ordinary least squares estimator of α^* , which is $\sqrt{T}$ -consistent. Although the number of parameters is strictly less than the sample size in this setting, the estimator is high-dimensional in the sense that the number of parameters equals N , which diverges as the sample size increases. Therefore, this case differs from the classical low-dimensional setting considered in Zou (2006). The weighted regularization has the following obvious property: when $\hat{\alpha}_i^{fe}$ is small, the penalty on the individual fixed effect α_i^* is large, and vice versa. Asymptotically, as $T \rightarrow \infty$, individuals with $\alpha_i^* = 0$ will receive an infinitely large penalty in the minimization, and thus $\hat{\alpha}_i = 0$. This property ensures the theoretical advantages of the weighted regularization.

To gain more insights on the weighted penalty, we compare it with other types of penalty functions. We first consider the unweighted ℓ_1 -penalty for a univariate parameter. This penalty constrains the absolute value of the estimator, and its non-differentiable point at zero enables the variable selection property. However, a drawback of the unweighted ℓ_1 -penalty is that it applies the same amount of penalty regardless of the true ℓ_1 -norm of the parameter. In other words, it tends to over-penalize large parameters, leading to downward bias in their estimates. A more reasonable penalty function is the smoothly clipped absolute deviation (SCAD) penalty described in Fan and Li (2001). The SCAD penalty behaves like the ℓ_1 -penalty near zero to preserve the variable selection property, but decreases in magnitude as a concave function as the size of the parameter increases, thereby reducing shrinkage bias. The SCAD penalty has a theoretical advantage over the unweighted ℓ_1 -penalty: small parameters are shrunk to zero, while the penalty on large parameters is bounded by construction. However, SCAD is computationally challenging, as the associated optimization problem is non-convex. As a result, there is no theoretical guarantee of convergence for algorithms used to solve it. The connection between this discussion and the current context is as follows: we can view the weighted ℓ_1 -penalty as a linear approximation of the SCAD penalty, which is inherently non-convex. In this sense, the weighted ℓ_1 -penalty achieves a similar reduction in shrinkage bias for large parameters, while maintaining the computational tractability of convex optimization, as in the unweighted ℓ_1 -penalty. This interpretation of weighted regularization as a linear approximation is discussed in more detail in Zou and Li (2008).

One final remark about the construction of the weights is that it is possible to update the weights iteratively following the steps below. First, estimate the fixed effects estimator $\hat{\beta}^{fe}$ and compute $\hat{\alpha}^{fe}$. Define the initial weights as $w_i^{(1)} = 1/|\hat{\alpha}_i^{fe}|$, and compute $\hat{\beta}^{(1)}$ and $\hat{\alpha}^{(1)}$ as the minimizers of equation (4) using weights $w_i^{(1)}$. For $k \geq 2$, define $w_i^{(k)} = 1/|\hat{\alpha}_i^{(k-1)}|$, and compute $\hat{\beta}^{(k)}$ and $\hat{\alpha}^{(k)}$ as the minimizers of equation (4) using weights $w_i^{(k-1)}$. In practice, the iteratively updated weights tend to produce increasingly sparse solutions, which we find redundant in the current context. Therefore, this direction will not be further pursued in the following discussion.

2.3 Penalty Selection

It is clear that the asymptotic behavior of the regularized estimator depends on the rate of the tuning parameter. Therefore, we index the tuning parameter λ_n by the sample size $n = NT$. In

practice, the optimal choice of λ_n for desirable asymptotic properties is infeasible, and the tuning parameter is typically selected using resampling techniques aimed at optimizing out-of-sample prediction performance.

The method recommended in this paper is a variation of the k -fold cross-validation approach. Recall that for a general estimation procedure that depends on a tuning parameter λ , the k -fold cross-validation can be implemented as follows: we first divide the data into k roughly equal-sized folds. Each fold is used once as a test set to evaluate the performance of the estimator, which is trained on the remaining $k - 1$ folds. This process is repeated k times—once for each fold—and the test errors are averaged across folds. The optimal tuning parameter is then selected to minimize this average test error. However, this procedure breaks down in the current context due to the lack of variation in the individual indicator matrix $\mathbf{D}$ within subsamples. Recall that $\mathbf{D}$ is an $NT \times N$ matrix, where the i th column of $\mathbf{D}$ equals 1 only for the rows corresponding to individual i and zero elsewhere. As a result, if individual i is excluded from a given subsample, the entire i th column of $\mathbf{D}$ becomes zero within that subsample, rendering the estimation ill-posed.

To address this challenge, we exploit the structure of panel data—where repeated observations are available for each individual over multiple periods—and propose a leave-one-period-out cross-validation approach. As the name suggests, for panel data containing observations on N individuals over T periods, we divide the data into T folds. In each fold, we leave out one of the T periods as the test set for all individuals. This subsampling strategy ensures that every individual is represented in all T folds, thereby avoiding the issue of unobserved fixed effects in the subsamples. The rest of the procedure follows the standard cross-validation scheme. This method is used in the numerical studies presented in the later sections.

We point out a few limitations of the proposed leave-one-period-out cross-validation method. First, the method works best for balanced panel data. Although it is possible to apply the same strategy to imbalanced panels by allowing repeated observations in different subsamples, the resulting estimate of the test error is generally less reliable. Second, the method does not extend to models with multidimensional fixed effects, such as the widely used two-way fixed effects model. Third, when the idiosyncratic errors $\varepsilon_{i1}, \dots, \varepsilon_{iT}$ exhibit dependence across T , the method tends to underestimate the test error and may lead to overfitting. However, this last issue is of less concern in the present context, where the primary goal is valid inference on β^* , rather than prediction.

Besides resampling techniques, model tuning methods that rely on information criteria—such as the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC)—can also be used in practice, although they lack theoretical justification in the current framework. For an application that uses BIC to select the tuning parameter for fixed effects in the context of quantile regression, see [Galvao and Montes-Rojas \(2010\)](#).

2.4 Computation

As mentioned earlier, a key advantage of incorporating weighted regularization into the fixed effects estimator—rather than using unweighted regularization—is that it reduces the shrinkage bias induced by regularization, while retaining the computational efficiency of standard ℓ_1 -regularization. Here, we describe how to transform equation (5) into an equivalent minimization problem that can be readily solved using the `glmnet` package, available in both R and python

(Friedman et al., 2010).

For a given tuning parameter and the weight matrix $\mathbf{W}$ defined previously, let $\mathbf{I}_p$ denote the p -dimensional identity matrix, and let $\mathbf{O}$ denote a zero matrix of appropriate dimension. Define the matrix

$$\mathbf{T} = \begin{bmatrix} \mathbf{I}_p & \mathbf{O}_{p \times N} \\ \mathbf{O}_{N \times p} & \mathbf{W} \end{bmatrix}, \quad (7)$$

and the associated change of variables $(\tilde{\beta}, \tilde{\alpha}) = \mathbf{T}(\beta, \alpha)$. Using this transformation, the objective function in equation (5) can be rewritten as

$$\left\| \mathbf{y} - \begin{bmatrix} \mathbf{X} & \mathbf{D} \end{bmatrix} \mathbf{T}^{-1} \begin{bmatrix} \tilde{\beta} \\ \tilde{\alpha} \end{bmatrix} \right\|_2^2 + \lambda \|\tilde{\alpha}\|_1. \quad (8)$$

By standard results from optimization theory, the minimizer of equation (8) corresponds to the minimizer of equation (5) after applying $\mathbf{T}^{-1}$. Since the upper-left block of $\mathbf{T}$ is the identity matrix and $\mathbf{T}$ is diagonal, we can simplify the computation. In particular, we can directly compute $\hat{\beta}$ by solving the following minimization problem:

$$\left\| \mathbf{y} - \mathbf{X}\beta - \mathbf{D}\mathbf{W}^{-1}\alpha \right\|_2^2 + \lambda \|\alpha\|_1. \quad (9)$$

Note that equation (9) is a standard lasso-type minimization problem with covariates $\mathbf{X}$ and $\mathbf{D}\mathbf{W}^{-1}$, except that β is excluded from the penalty. This formulation can be directly handled by `glmnet`. Furthermore, there is no need to transform the solution back using $\mathbf{T}^{-1}$ if the primary interest lies in estimating β^* .

The `glmnet` package implements the Coordinate Descent Algorithm (CDA) (Tseng, 2001), which exploits the additive structure of the ℓ_1 -regularization. The computational complexity is equivalent to that of ordinary least squares and can scale efficiently to problems with a large number of observations. We note that CDA was originally written for optimization problems in which all parameters are penalized. However, the modification to accommodate partially penalized minimization is straightforward, and the details are omitted here. In `glmnet`, excluding β from regularization is achieved by adjusting the penalty weights.

3 Asymptotic Analysis

3.1 Notations and Conventions

In this section, we present the main results of the paper. We first introduce some notations and conventions that will be used throughout this section.

We maintain the assumption of the linear panel model with one-way fixed effects specified in equation (1), where β^* and α^* are the true parameters of dimensions p and N , respectively. We define the matrices $\mathbf{y}$, $\mathbf{X}$, $\mathbf{D}$, and ε by stacking the variables y_{it} , $\mathbf{x}_{it}'$, $\mathbf{d}_i'$, and ε_{it} first across t , then across i . Let $n = NT$ to be the total number of observations. Instead of working with the weights proposed in the previous section, we consider a general weight vector $\mathbf{w} = (w_1, \dots, w_N)'$ that satisfies the conditions stated in the upcoming assumptions.

We follow the convention in the literature and let $s := \{i: \alpha_i^* \neq 0\}$ denote the support of α^* , i.e., the set of indices corresponding to the nonzero entries of α^* . The set s depends implicitly

on N , and with a slight abuse of notation, we let $s_N := |s|$ denote its cardinality—representing the number of individuals with nonzero fixed effects, or equivalently, the number of nonzero elements in α^* . Similarly, let $z := s^c$ be the complement of s , and define $z_N := |z|$ as its cardinality, i.e., the number of zero entries in α^* . Without loss of generality, we assume that $s = \{1, \dots, s_N\}$ and $z = \{s_N + 1, \dots, s_N + N\}$.

We decompose the $N \times 1$ vector α^* into an $s_N \times 1$ vector α_s^* , which contains only the nonzero coordinates of α^* , and a $z_N \times 1$ zero vector $\alpha_z^* = \mathbf{0}$, which contains only the zero coordinates of α^* . For convenience, we sometimes identify the $s_N \times 1$ vector α_s^* with an $N \times 1$ vector whose j th coordinate equals α_j^* if $j \in s$, and 0 if $j \in z$. Under this convention, the product $\mathbf{D}\alpha_s^* = \mathbf{D}\alpha^*$ is well-defined. In the same spirit, we also decompose the $NT \times N$ matrix $\mathbf{D}$ into an $NT \times s_N$ matrix $\mathbf{D}_s$, which contains only the columns of $\mathbf{D}$ indexed by s , and an $NT \times z_N$ matrix $\mathbf{D}_z$, which contains only the columns of $\mathbf{D}$ indexed by z . We note that this method of decomposition using subscripts s and z extends to arbitrary vectors and matrices of compatible dimensions.

Let $\mathbf{a} = (a_1, \dots, a_m)'$ and $\mathbf{b} = (b_1, \dots, b_m)'$ be real-valued vectors in $\mathbb{R}^m$. Let $\|\mathbf{a}\|_1$, $\|\mathbf{a}\|_2$, and $\|\mathbf{a}\|_{\psi^2}$ denote the ℓ_1 -norm, ℓ_2 -norm, and the Orlicz norm associated with the function $\psi_2(x) = e^{x^2} - 1$, respectively. Define $|\mathbf{a}| := (|a_1|, \dots, |a_m|)'$ and $\text{sgn}(\mathbf{a}) := (\text{sgn}(a_1), \dots, \text{sgn}(a_m))'$, with the convention that $\text{sgn}(0) = 0$. The inequality $\leq$ between two vectors is understood coordinate-wise; that is, $\mathbf{a} \leq \mathbf{b}$ if and only if $a_i \leq b_i$ for all $1 \leq i \leq m$. Similarly, for a scalar $c \in \mathbb{R}$, we write $\mathbf{a} \leq c$ if and only if $a_i \leq c$ for all $1 \leq i \leq m$. For $a, b \in \mathbb{R}$, we write $a \lesssim b$ if and only if $a \leq cb$ for some universal constant $c \geq 0$.

For an $n \times m$ matrix $\mathbf{A}$, let $[\mathbf{A}]_j$ denote the j th column of $\mathbf{A}$, viewed as an $n \times 1$ vector in $\mathbb{R}^n$. When $\mathbf{A}$ is referring to the covariates matrix $\mathbf{X}$ or the individual indicator covariates $\mathbf{D}$, we will drop the bracket and use $\mathbf{X}_j$ or $\mathbf{D}_i$ to denote the j th or i th column of the matrix $\mathbf{X}$ and $\mathbf{D}$ respectively. Let $\|\mathbf{A}\|_2$ denote the ℓ_2 operator norm of $\mathbf{A}$. Let $\Lambda^{\min}(\mathbf{A})$ and $\Lambda^{\max}(\mathbf{A})$ denote the smallest and largest eigenvalues of $\mathbf{A}$, respectively, when they are well-defined. We denote by $\mathbf{P}_\mathbf{A} := \mathbf{A}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'$ the projection matrix onto the column space of $\mathbf{A}$. Similarly, the projection matrix onto the orthogonal complement of the column space of $\mathbf{A}$ is denoted by $\mathbf{R}_\mathbf{A} := \mathbf{I} - \mathbf{P}_\mathbf{A}$, where $\mathbf{I}$ is the identity matrix of appropriate dimension.

3.2 Convergence Rate

In this subsection, we derive a non-asymptotic ℓ_2 error bound for the sparse fixed effects estimators $\hat{\beta}$ and $\hat{\alpha}$ defined in the previous section, under the following assumptions:

A1: Let $\mathbf{w} = (w_1, \dots, w_N)'$ be the predetermined data-dependent weights used in equation (4), and let $w_z^{\min} := \min_{i \in z} w_i$. Assume that

$$1 \leq \lambda_n w_z^{\min}, \quad \sqrt{p + s_N} \leq \lambda_n \|\mathbf{w}_s\|_2, \quad \text{and} \quad \|\mathbf{w}_s\|_2 \leq \sqrt{s_N} w_z^{\min}. \quad (10)$$

A2: Assume $\mathbf{X}$ and $\mathbf{D}$ are fixed. Define the augmented covariate matrices and the empirical Gram matrix as

$$\mathbf{C} := \begin{bmatrix} \mathbf{X} & \mathbf{D} \end{bmatrix}, \quad \mathbf{C}_s := \begin{bmatrix} \mathbf{X} & \mathbf{D}_s \end{bmatrix}, \quad \text{and} \quad \Sigma := \frac{1}{n} \mathbf{C}'\mathbf{C}. \quad (11)$$

Assume there exists a constant $\phi > 0$ such that

$$\phi^2 \leq \frac{\gamma' \Sigma \gamma}{\|\gamma_s\|_2^2}, \quad (12)$$

for all $(p + N) \times 1$ vectors γ satisfying the condition $\|\gamma_{s^c}\|_1 \leq 3\sqrt{s_N}\|\gamma_s\|_2$, where γ_s consists of the first p coordinates of γ together with the coordinates indexed by $p + i$ for $i \in s$.

Theorem 1 (Convergence Rate). Assume **A1** and **A2**. Let $\hat{\gamma} = (\hat{\beta}', \hat{\alpha}')'$, $\hat{\gamma}_s = (\hat{\beta}', \hat{\alpha}'_s)'$, $\gamma^* = ((\beta^*)', (\alpha^*)')'$, and $\gamma_s^* = ((\beta^*)', (\alpha_s^*)')'$. Define the event

$$\mathcal{R} := \left\{ \max_{1 \leq j \leq p} \frac{2}{n} |\varepsilon' \mathbf{X}_j| \leq M_0 \quad \text{and} \quad \max_{1 \leq i \leq N} \frac{2}{n} |\varepsilon' \mathbf{D}_i| \leq M_0 \right\}, \quad (13)$$

for some constant $M_0 \leq \frac{1}{2}$. Then, on the event $\mathcal{R}$, the following error bound holds:

$$\sum_{i \in z} w_i |\hat{\alpha}_i| + \|\mathbf{w}_s\|_2 \|\hat{\gamma}_s - \gamma_s^*\|_2 \leq \frac{2\lambda_n \|\mathbf{w}_s\|_2^2}{\phi^2}. \quad (14)$$

Remark 1. Inequality (14) provides convergence rate for the significant parameters in the non-asymptotic form. It is interesting compared to non-asymptotic error bounds for other penalized estimators in the sense that one would typically expect the upper bound to depend on the growth rate of s_N . For a discussion of non-asymptotic error bounds for the standard Lasso estimator, see Chapter 2 of [Wainwright \(2019\)](#). Here, by incorporating weights, we are able to bound the ℓ_1 error by $\|\mathbf{w}_s\|_2$, which is a consequence of Assumption **A1**. For consistency, a sufficient condition is that $\lambda_n \|\mathbf{w}_s\|_2 \rightarrow 0$, which can be ensured by a carefully constructed weighting scheme. In this way, the estimation problem is essentially transformed into finding suitable weights—something that is relatively straightforward in the current context.

Remark 2. By introducing the event $\mathcal{R}$, we separate the deterministic and stochastic components of the problem, with the latter being controlled by $\mathcal{R}$. The advantage of this approach is that the error bound becomes independent of the stochastic structure of the errors ε_{it} . To ensure that $\mathbb{P}(\mathcal{R}) \rightarrow 1$, it suffices to show that the tail probability of the maximum of a finite collection of random variables converges to zero. Techniques and conditions for proving such results can be found in Chapter 2 of [Wainwright \(2019\)](#). Indeed, in the proof of the next theorem, we provide a concrete example of how to establish the convergence of probability for this type of tail event. Also note that when ε_{it} are i.i.d. Gaussian, the conditions required for $\mathbb{P}(\mathcal{R}) \rightarrow 1$ are satisfied.

Remark 3. Assumption **A1** imposes mild restrictions on the weights $\mathbf{w}$, and it should be satisfied by any reasonable choice of weights. Assumption **A2** is a version of the Restricted Eigenvalue (RE) condition, adapted to the current setting to ensure identification of the parameters. This type of assumption is standard in the literature on error bounds for penalized estimators. Assumption **A2** is implied by a lower bound on the smallest eigenvalue of the empirical Gram matrix Σ . It is also possible to express this condition in terms of the population Gram matrix, in which case Theorem 1 could be generalized for random covariates. However, this extension is mathematically of the same nature as the fixed covariate case and is therefore not pursued here. For a detailed discussion on the relationships among various versions of the RE condition, see [van de Geer and Bühlmann \(2009\)](#).

Remark 4. One could argue that the condition relying on the exact number of nonzero coefficients s_N is too restrictive, and that it would be desirable for the theory to accommodate cases where many α_i^* are small but nonzero—for example, $\alpha_i^* \sim \mathcal{N}(0, 1)$. This generalization can be achieved by defining an oracle parameter

$$a^* := \operatorname{argmin}_{\alpha \in \mathbb{R}^N} \left\{ \|\mathbf{D}\alpha^* - \mathbf{D}\alpha\|_2^2 + \lambda \|\alpha\|_0 \right\}, \quad (15)$$

where $\lambda \geq 0$ is a ‘population-level’ regularization parameter that governs the trade-off between approximation error and sparsity. A convergence result similar to Theorem 1 can be established with α^* replaced by a^* , and the corresponding error bound includes an additional term, $\|\mathbf{D}\alpha^* - \mathbf{D}a^*\|_2^2$, to account for the regularization error. Then, all assumptions that restrict the growth rate of s_N can be replaced by analogous assumptions on the growth rate of $\|a^*\|_0$.

3.3 Oracle Property

In this subsection, we first show that when the true α_i^* equals 0, it is estimated as 0 with probability tending to one; that is, $\hat{\alpha}_z \rightarrow \mathbf{0}$ with probability tending to one. In fact, we prove that $\text{sgn}(\hat{\alpha}) = \text{sgn}(\alpha^*)$ with probability tending to one, which is a stronger property. Furthermore, we show that β^* , along with the nonzero components of α^* , are estimated as accurately as if the true support set s of α^* were known in advance. That is, $\hat{\beta}$ and $\hat{\alpha}_s$ converge in distribution to a normal distribution with the same asymptotic variance as would be obtained if the individual indicator matrix $\mathbf{D}_s$ —rather than the full $\mathbf{D}$ —were observed.

For the results in this subsection, we impose the following assumptions:

B1: Assume that $\mathbf{X}$ is fixed. For $1 \leq i \leq N$, $1 \leq t \leq T$, let ε_{it} be i.i.d. random errors with mean zero and variance $\sigma^2 < \infty$. Furthermore, assume that the tail probabilities of ε_{it} decay exponentially; that is, for some constants $A > 0$ and $B > 0$,

$$\mathbb{P}(|\varepsilon_{it}| > v) \leq A \exp\{-Bv^2\}, \quad (16)$$

for all $v > 0$.

B2: There exists a constant $c^{\min} > 0$ such that

$$c^{\min} \leq \Lambda^{\min} \left(\frac{1}{n} \mathbf{D}_s' \mathbf{R}_X \mathbf{D}_s \right) \quad (17)$$

holds uniformly over all sample sizes $n = NT$.

B3: Let $\mathbf{w} = (w_1, \dots, w_N)'$ be the predetermined data-dependent weights. Define $a_N^{\min} := \min_{i \in s} \alpha_i^*$ and $a_N^{\max} := \max_{i \in s} \alpha_i^*$. We assume that $\mathbf{w}$ is r_n -consistent in the sense that

$$\mathbb{P}(\mathbf{w}_s > 1/a_N^{\min}) \rightarrow 0 \quad \text{and} \quad \mathbb{P}(\mathbf{w}_z < r_n) \rightarrow 0, \quad (18)$$

where $r_n \rightarrow \infty$.

B4: Assume the following sequences converge to zero:

$$\frac{\sqrt{\log s_N}}{\sqrt{NT} a_N^{\min}}, \quad \frac{\lambda_n \sqrt{s_N}}{NT(a_N^{\min})^2}, \quad \frac{\sqrt{NT \log z_N}}{\lambda_n r_n}, \quad \frac{s_N^2}{r_n N a_N^{\min}} \rightarrow 0. \quad (19)$$

B5: Let $\Sigma_{11} := \frac{1}{n} \mathbf{C}_s' \mathbf{C}_s$. We assume

$$\frac{\Lambda^{\max}(\Sigma_{11})}{\Lambda^{\min}(\Sigma_{11})} = O(1) \quad \text{and} \quad \frac{1}{NT \Lambda^{\min}(\Sigma)} \max_{1 \leq i \leq N, 1 \leq t \leq T} \{\mathbf{x}_{it}' \mathbf{x}_{it} + T\} \rightarrow 0. \quad (20)$$

Theorem 2 (Variable Selection Consistency). Under assumptions **B1–B4**, we have

$$\mathbb{P}(\text{sgn}(\hat{\alpha}) = \text{sgn}(\alpha^*)) \rightarrow 1. \quad (21)$$

Remark 5. Theorem 2 establishes variable selection consistency. It can be viewed as a strengthened version of the result in Zhao and Yu (2006), improved through the incorporation of weights in the estimation procedure. This result also represents a variation of the proposition in Huang et al. (2008b), adapted to the context of panel data with partial regularization.

Theorem 3 (Asymptotic Normality). Let $\mathbf{c}'_{sit}$ denote the it th row of $\mathbf{C}_s$. Define $v_n^2 := \sigma^2 \mathbf{u}' \Sigma_{11}^{-1} \mathbf{u}$, where $\mathbf{u}$ is a $(p + s_N) \times 1$ vector satisfying $\|\mathbf{u}\|_2 \leq 1$. Under assumptions **B1–B5**, we have

$$\sqrt{NT} v_n^{-1} \mathbf{u}' (\hat{\gamma}_s - \gamma_s^*) = \sum_{i=1}^N \sum_{t=1}^T \frac{1}{\sqrt{NT}} v_n^{-1} \mathbf{u}' \Sigma_{11}^{-1} \mathbf{c}_{sit} \varepsilon_{it} + o_p(1) \xrightarrow{d} \mathcal{N}(0, 1). \quad (22)$$

Remark 6. Taken together, Theorems 2 and 3 establish the oracle property of the proposed estimators $\hat{\beta}$ and $\hat{\alpha}$. For a penalized estimator that performs both model selection and estimation simultaneously, the oracle property refers to the ability to correctly identify non-significant parameters with probability tending to one, while estimating the significant parameters as if their locations were known in advance (Fan and Li, 2001). Theorems 2 and 3 extend the oracle property of the high-dimensional adaptive Lasso (Huang et al., 2008b) to the context of fixed effects models.

Remark 7. Theorem 3 justifies the intuition that the efficiency of the fixed effects estimator can be improved through weighted regularization. In finite samples, the sparse fixed effects estimator outperforms the classical fixed effects estimator via a variance–bias trade-off. This phenomenon will be demonstrated in the upcoming numerical studies, where estimator performance is evaluated using an empirical risk function under squared error loss. In large samples, however, the source of efficiency gain relative to the classical estimator is fundamentally different. Theorem 3 implies that the sparse fixed effects estimator is asymptotically unbiased. Thus, its superior efficiency arises entirely from variance reduction. This variance reduction, in turn, results from the oracle property—specifically, the estimator’s ability to recover the true underlying model with reduced model complexity, that is, fewer significant coefficients to estimate.

Remark 8. Unfortunately, Assumption **B1** excludes many interesting examples in econometrics, such as dynamic panel models. This is the price we pay for allowing $s_N \rightarrow \infty$. A random variable with exponentially decaying tail probabilities is called sub-Gaussian, and this property can be characterized by several equivalent conditions. Note that the normal distribution satisfies this definition. Sub-Gaussianity is a natural assumption when establishing convergence of tail bounds, such as the probability of the event described in Theorem 1. Assumption **B3** imposes a crude but necessary restriction on the weights. Assumption **B2** is more likely to hold when s_N is small, as it concerns the eigenvalue condition for a submatrix of the design. Finally, Assumption **B4** is only needed for Theorem 3, where it is used to verify the Lindeberg–Feller condition for the central limit theorem.

4 Simulation Study

In this section, we report the simulation results for the Sparse Fixed Effects estimators $\hat{\beta}$ and $\hat{\alpha}$ defined previously. Our primary focus is on estimation efficiency, measured by the mean squared error (MSE). We compare the performance of the proposed estimator to several classical estimators commonly used in econometrics, including the random effects estimator and the fixed effects estimator, as well as to standard machine learning approaches such as the unweighted Lasso and

the post-selection estimator. The oracle estimator, which assumes knowledge of the true support, is used as a benchmark.

We simulate data from the following linear model:

$$y_{it} = \beta_1^* x_{1it} + \beta_2^* x_{2it} + s \alpha_i^* + \varepsilon_{it}, \quad (23)$$

where

$$s \sim \text{Bernoulli}(\psi), \quad x_{1it} = \rho \alpha_i^* + (1 - \rho) \mathcal{N}(0, 1), \quad \alpha_i^* \sim \mathcal{N}(1, 1), \quad x_{2it}, \varepsilon_{it} \sim \mathcal{N}(0, 1). \quad (24)$$

We let $\psi \in \{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ control the sparsity of the fixed effects, where $\psi = 0$ implies that no individual has nonzero fixed effects, and $\psi = 1$ implies that all individuals have fixed effects drawn from a standard normal distribution. We also vary $\rho \in \{0, 0.2, 0.4, 0.6, 0.8\}$ to control the degree of endogeneity of the treatment variable x_{1it} : for example, $\rho = 0$ corresponds to the exogenous case, while $\rho = 0.8$ represents a highly endogenous scenario. Note that ρ understates the sample correlation between α_i and x_{1it} , since α_i has no variation along the time dimension. For this simulation study, we set $N = 100$ and $T = 10$, and we repeat the simulation 10,000 times.

We emphasize that the purpose of this data-generating process is not to mimic any real-world dataset, but rather to illustrate how the sparsity and endogeneity of the data interact with the performance of the Sparse Fixed Effects estimator. Results for alternative data-generating processes are reported in the appendix.

We focus on the estimation of $\hat{\beta}_1$. Table 1 reports the mean and standard deviation of the estimator $\hat{\beta}_1$. As expected, both bias and standard deviation increase with the sparsity parameter ψ and the endogeneity parameter ρ . Table 2 compares the probabilities of correctly identifying zero and nonzero coefficients using the weighted Lasso versus the standard Lasso. The results show that the weighted Lasso substantially improves variable selection performance. Although our theoretical results assume a consistent first-stage estimator for the fixed effects, this simulation suggests that $T = 10$ is sufficient in practice to achieve good performance. Table 3 compares the Sparse Fixed Effects estimator with the classical fixed effects estimator using empirical risk as the evaluation metric. The results demonstrate that the proposed estimator outperforms the classical fixed effects estimator when $\psi < 0.6$. This finding is consistent with our theoretical requirement that the number of nonzero fixed effects must grow at a controlled rate. Finally, we observe that the efficiency gains from the Sparse Fixed Effects estimator increase with higher values of ρ . This suggests that the estimator performs particularly well when the treatment variable is highly correlated with unobserved individual-specific heterogeneity.

Figures 1 and 2 compare the empirical distributions of the Sparse Fixed Effects (SFE) estimator with those of other commonly used methods. The endogeneity parameter is fixed at $\rho = 0.4$, while the sparsity parameter ψ is varied. Additional results and alternative data-generating processes are presented in the appendix.

Table 1: Mean (Standard Deviation)

	$\rho = 0.0$	$\rho = 0.2$	$\rho = 0.4$	$\rho = 0.6$	$\rho = 0.8$
$\psi = 0.0$	1 (0.02)	1 (0.02)	1 (0.02)	1 (0.02)	1 (0.02)
$\psi = 0.1$	1 (0.02)	1 (0.02)	1 (0.02)	1 (0.02)	1 (0.02)
$\psi = 0.2$	1 (0.02)	1 (0.02)	1.01 (0.02)	1.01 (0.02)	1.01 (0.02)
$\psi = 0.3$	1 (0.02)	1 (0.02)	1.01 (0.02)	1.01 (0.02)	1.01 (0.02)
$\psi = 0.4$	1 (0.02)	1 (0.02)	1.01 (0.02)	1.02 (0.02)	1.02 (0.03)
$\psi = 0.5$	1 (0.02)	1 (0.02)	1.01 (0.02)	1.02 (0.03)	1.03 (0.03)
$\psi = 0.6$	1 (0.02)	1.01 (0.02)	1.02 (0.03)	1.03 (0.03)	1.04 (0.04)
$\psi = 0.7$	1 (0.02)	1.01 (0.02)	1.02 (0.03)	1.03 (0.03)	1.05 (0.05)
$\psi = 0.8$	1 (0.02)	1.01 (0.02)	1.02 (0.03)	1.04 (0.04)	1.07 (0.06)
$\psi = 0.9$	1 (0.02)	1.01 (0.02)	1.02 (0.03)	1.04 (0.04)	1.10 (0.08)
$\psi = 1.0$	1 (0.02)	1 (0.02)	1.02 (0.03)	1.04 (0.04)	1.13 (0.14)

Notes: This table reports the mean and the standard deviation of the sparse fixed effect estimator $\hat{\beta}_1$ with $\beta^* = 1$.

Table 2: Selection Consistency Rate of SFE

	$\rho = 0.0$	$\rho = 0.2$	$\rho = 0.4$	$\rho = 0.6$	$\rho = 0.8$
$\psi = 0.0$	0.88 (0.97)	0.88 (0.97)	0.88 (0.97)	0.88 (0.97)	0.89 (0.97)
$\psi = 0.1$	0.85 (0.81)	0.85 (0.81)	0.85 (0.81)	0.85 (0.80)	0.85 (0.80)
$\psi = 0.2$	0.84 (0.72)	0.84 (0.72)	0.84 (0.71)	0.83 (0.70)	0.83 (0.69)
$\psi = 0.3$	0.83 (0.68)	0.83 (0.68)	0.83 (0.66)	0.83 (0.64)	0.82 (0.61)
$\psi = 0.4$	0.83 (0.67)	0.83 (0.67)	0.83 (0.65)	0.82 (0.61)	0.82 (0.56)
$\psi = 0.5$	0.83 (0.68)	0.83 (0.68)	0.83 (0.66)	0.83 (0.62)	0.82 (0.57)
$\psi = 0.6$	0.84 (0.71)	0.84 (0.71)	0.84 (0.69)	0.84 (0.66)	0.83 (0.62)
$\psi = 0.7$	0.86 (0.76)	0.86 (0.76)	0.85 (0.75)	0.85 (0.72)	0.84 (0.71)
$\psi = 0.8$	0.87 (0.83)	0.87 (0.82)	0.87 (0.82)	0.87 (0.81)	0.86 (0.80)
$\psi = 0.9$	0.90 (0.90)	0.90 (0.90)	0.90 (0.90)	0.90 (0.90)	0.90 (0.90)
$\psi = 1.0$	0.94 (0.99)	0.94 (0.99)	0.94 (0.99)	0.95 (1.00)	0.95 (1.00)

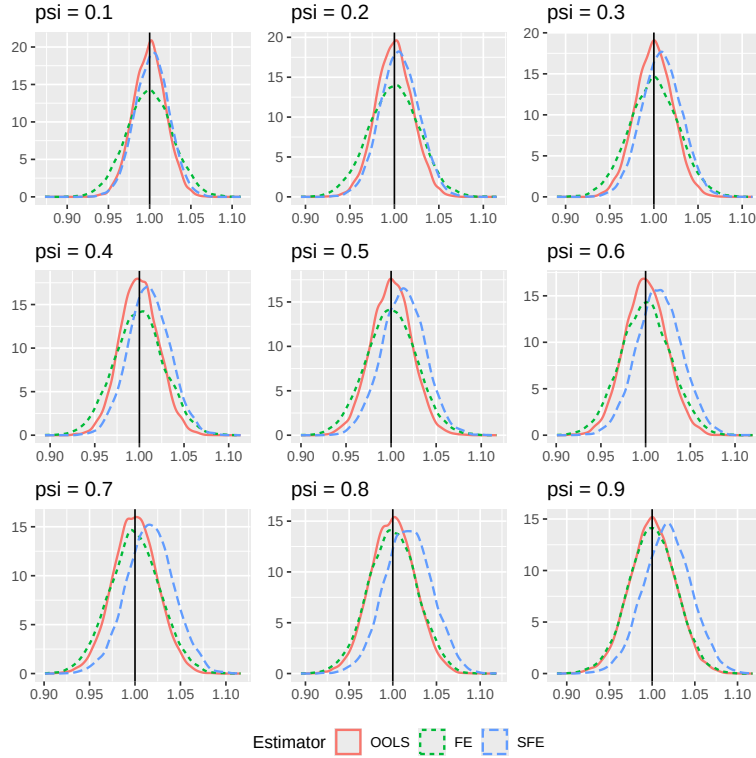
Notes: This table reports the probabilities of correctly identifying the zero and nonzero coefficients of the Sparse Fixed Effects (SFE) estimator under different values of ψ and ρ . The corresponding probabilities for the standard Lasso are reported in parentheses. Values highlighted in **red** indicate that the SFE outperforms the standard Lasso, while values in **blue** indicate equal performance.

Table 3: Empirical Risk of SFE

	$\rho = 0.0$	$\rho = 0.2$	$\rho = 0.4$	$\rho = 0.6$	$\rho = 0.8$
$\psi = 0.0$	0.025 (0.028)	0.036 (0.043)	0.040 (0.076)	0.033 (0.173)	0.027 (0.674)
$\psi = 0.1$	0.026 (0.028)	0.038 (0.044)	0.044 (0.078)	0.041 (0.176)	0.037 (0.688)
$\psi = 0.2$	0.026 (0.028)	0.039 (0.044)	0.051 (0.080)	0.053 (0.177)	0.052 (0.701)
$\psi = 0.3$	0.026 (0.027)	0.040 (0.043)	0.059 (0.078)	0.068 (0.172)	0.072 (0.679)
$\psi = 0.4$	0.027 (0.028)	0.042 (0.044)	0.066 (0.079)	0.092 (0.174)	0.110 (0.707)
$\psi = 0.5$	0.027 (0.028)	0.043 (0.044)	0.077 (0.078)	0.120 (0.177)	0.165 (0.697)
$\psi = 0.6$	0.028 (0.028)	0.044 (0.043)	0.087 (0.079)	0.166 (0.177)	0.272 (0.693)
$\psi = 0.7$	0.027 (0.028)	0.045 (0.043)	0.096 (0.078)	0.219 (0.174)	0.470 (0.679)
$\psi = 0.8$	0.028 (0.028)	0.048 (0.044)	0.103 (0.077)	0.286 (0.173)	0.892 (0.699)
$\psi = 0.9$	0.027 (0.027)	0.047 (0.043)	0.110 (0.078)	0.339 (0.174)	1.583 (0.698)
$\psi = 1.0$	0.028 (0.028)	0.045 (0.043)	0.106 (0.077)	0.365 (0.172)	3.599 (0.709)

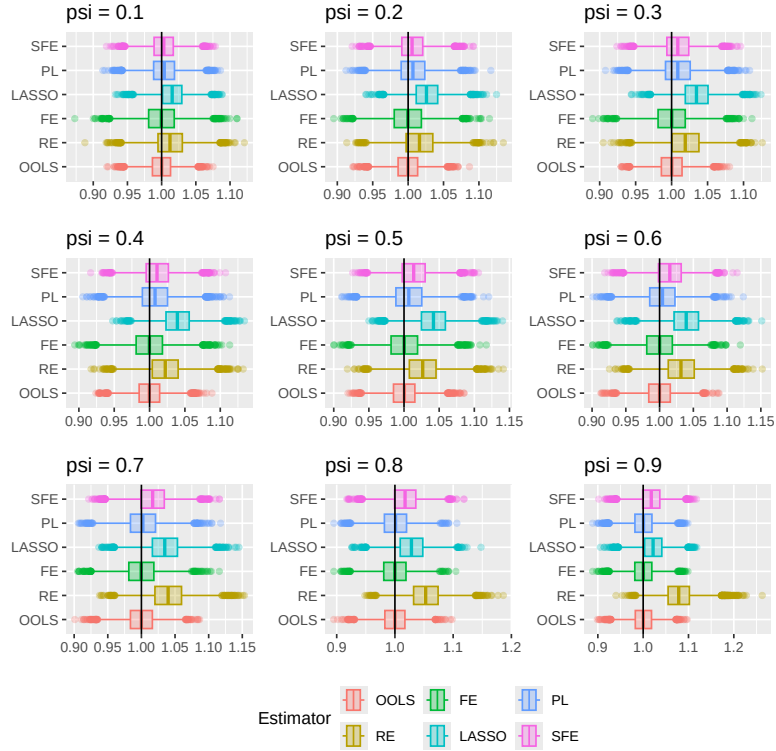
Notes: This table reports the empirical risk ($\times 100$) of the Sparse Fixed Effects (SFE) estimator under different values of ψ and ρ . The empirical risks of the classical fixed effects (FE) estimator are reported in parentheses. Values highlighted in red indicate that the SFE outperforms the classical FE estimator, while values in blue indicate equal performance.

Figure 1: Distribution of OOLS/FE/SFE



Notes: This figure compares the empirical distributions of the Oracle Ordinary Least Squares (OOLS) estimator, the classical fixed effects estimator (FE), and the Sparse Fixed Effects (SFE) estimator. We fix $\rho = 0.4$ and vary the sparsity parameter ψ .

Figure 2: Distribution of OOLS/RE/FE/LASSO/PL/SFE



Notes: This figure presents box plots of the Oracle Ordinary Least Squares (OOLS) estimator, the Random Effects estimator (RE), the classical Fixed Effects estimator (FE), the standard Lasso, the post-selection Lasso (PL), and the Sparse Fixed Effects estimator (SFE). We fix $\rho = 0.4$ and vary the sparsity parameter ψ .

5 Data Example

In this section, we revisit the study by [Donohue III and Levitt \(2001\)](#) on the effect of legalized abortion on crime rates, using the dataset provided in the replication package of [Belloni et al. \(2014\)](#), to illustrate an empirical application of the proposed estimator.

We begin by briefly reviewing the empirical setting discussed in [Donohue III and Levitt \(2001\)](#). The paper outlines two channels through which the legalization of abortion may affect crime rates. First, increased access to abortion reduces the size of future cohorts, which simply leads to fewer individuals in the crime-prone age group (15–25 years) several years later. Second, and more subtly, the paper highlights that access to abortion gives women greater control over the timing of childbirth. This increased control may result in children being born into more favorable environments, which in turn could improve childhood development and reduce the likelihood of criminal behavior in adulthood. This insight aligns with findings in the education and psychology literature, which emphasize the long-term effects of early childhood environments on behavioral outcomes.

The basic empirical model in [Donohue III and Levitt \(2001\)](#) is specified as

$$Crime_{it} = Abortion_{it}\beta + \mathbf{x}'_{it}\gamma + \alpha_i + \delta_t + \varepsilon_{it}, \quad (25)$$

where i indexes states⁴, and t indexes years, ranging from 1985 to 1997. The term α_i captures state-specific, time-invariant heterogeneity, while δ_t represents time-specific aggregate shocks common across states. The vector $\mathbf{x}_{it}$ includes a set of covariates used to control for confounding factors⁵. Our primary interest lies in estimating γ , the causal effect of $Abortion_{it}$ —a measure of the abortion rate at the state level—on $Crime_{it}$, which represents the crime rate at the state level. We estimate the model separately for three categories of crime: violent crime, property crime, and murder. The treatment variable $Abortion_{it}$ is constructed as a weighted average of lagged abortion rates, where the weights reflect the proportion of each crime type committed by different age groups.⁶

We first estimate β using the within transformation. Specifically, we eliminate the state-specific fixed effects by demeaning the regressors at the state level and control for time-specific fixed effects by including year indicators from 1986 to 1997.⁷ Classical standard errors—assuming homoskedasticity and no serial correlation—are reported for the within transformation estimator.

Next, we compute the estimated state-specific fixed effects and use them to construct the weights required for estimating the Sparse Fixed Effects (SFE) estimator. Prior to estimation, all variables are mean-centered to impose the normalization condition

$$\sum_{i=1}^N \hat{\alpha}_i^{fe} = 0, \quad (26)$$

which has no effect on the estimation of the other coefficients. Figure 3 plots the distribution of the estimated fixed effects $\hat{\alpha}_i^{fe}$, calculated using the previously stated formula. As shown in the figure, after recentering the variables, the estimated fixed effects are approximately bell-shaped and centered around zero across all three outcome measures. This provides preliminary justification for implementing a regularized estimator.

The Sparse Fixed Effects estimator is then computed following the procedure described earlier. The tuning parameter λ is selected via the T -fold cross-validation method. Standard errors for the SFE estimator are computed using the following steps: (1) Estimate the SFE coefficients using the selected tuning parameter λ ; (2) Exclude covariates whose coefficient estimates are exactly zero; (3) Compute a consistent estimator of σ^2 from the residual sum of squares, adjusting for degrees of freedom equal to the sample size minus the number of regressors and minus the number of nonzero fixed effects selected by the SFE; (4) Apply the standard formula for computing standard errors, treating the model selected by the SFE as the true model. This procedure is justified by Theorem 3 and can be extended to accommodate heteroskedasticity-robust standard errors or standard errors clustered at the appropriate level.

Figure 3 compares the distribution of the estimated fixed effects from the classical Fixed Effects (FE) estimator and the Sparse Fixed Effects (SFE) estimator. In all three cases—violent crime, property crime, and murder—the SFE estimator yields slightly more concentrated estimates, as expected due to the regularization. Figure 4 presents scatter plots comparing the estimated fixed effects from the FE and SFE estimators. For violent crime and murder, the SFE estimates tend to be smaller in magnitude, reflecting the shrinkage property of the regularized estimator. However, in the case of property crime, the relationship is less clear.

⁴Here, the cross-sectional unit is states rather than individuals.

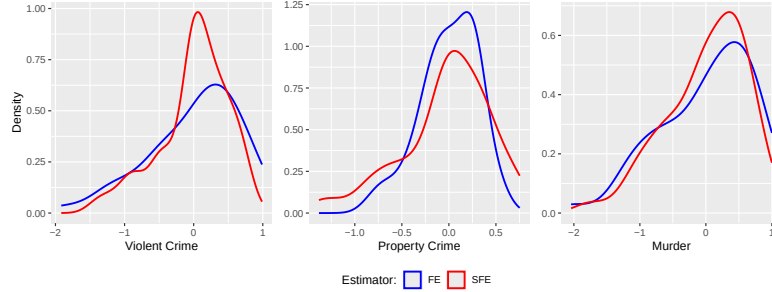
⁵The control vector $\mathbf{x}_{it}$ includes measures of prisoners per capita, police per capita, the unemployment rate, income, poverty rate, AFDC generosity, an indicator for concealed weapons laws, and beer consumption per capita.

⁶For example, if 60 percent of violent crime is committed by 18-year-olds and 40 percent by 19-year-olds in state i , then $Abortion_{it}$ is constructed as 0.6 times the abortion rate in state i at $t - 18$ plus 0.4 times the abortion rate at $t - 19$.

⁷The year 1985 is excluded for identification.

The estimated treatment effects of abortion and their corresponding standard errors are reported in Table 4. The estimates obtained using the proposed method are not qualitatively different from those of the classical estimator, although they exhibit slightly smaller standard errors.

Figure 3: Distribution of $\hat{\alpha}^{fe}$ and $\hat{\alpha}^{sfe}$



Notes: This figure presents the density plots of $\hat{\alpha}^{fe}$ and $\hat{\alpha}^{sfe}$. Note that $\sum_{i=1}^N \hat{\alpha}_i^{fe} = 0$ is a consequence of recentering the data and holds generally.

Figure 4: Comparing $\hat{\alpha}^{fe}$ and $\hat{\alpha}^{sfe}$



Notes: This figure presents scatter plots comparing $\hat{\alpha}^{fe}$ and $\hat{\alpha}^{sfe}$. For violent crime and murder, the shrinkage effect of the Sparse Fixed Effects (SFE) estimator is evident, as it tends to produce smaller estimates than the classical Fixed Effects (FE) estimator. While for property crime, the relationship is less clear.

Table 4: Estimated Effects of Abortion on Crime Rates

	Violent Crime		Property Crime		Murder	
	Estimate	SE	Estimate	SE	Estimate	SE
Within Transformation	-0.1351	0.0218	-0.0931	0.0104	-0.1336	0.0476
Sparse Fixed Effects	-0.1040	0.0215	-0.0553	0.0094	-0.1460	0.0385

Notes: This table compares the estimated effects of abortion on different categories of crime using the classical fixed effects estimator (within transformation) and the Sparse Fixed Effects (SFE) estimator proposed in this paper. Standard errors for the classical fixed effects estimator are computed under the assumption of homoskedasticity and no serial correlation. Standard errors for the SFE estimator are computed under the same assumptions, treating the selected model as the true model—a justification supported by the theoretical results established in this paper.

6 Conclusion and Extensions

In this paper, we explore the possibility of improving classical fixed effects estimation by incorporating ideas from the machine learning literature. We show that the proposed Sparse Fixed Effects (SFE) estimator can outperform the classical fixed effects estimator in certain settings, as supported by both asymptotic analysis and finite-sample simulations. A real-world empirical application further demonstrates the practical relevance of the method.

Several avenues for future research remain open:

1. [Kock \(2013\)](#) extended the oracle-efficient distribution for the Bridge estimator from [Huang et al. \(2008a\)](#) to a linear panel data model with random regressors and large- N asymptotics, under the assumption of a fixed parameter dimension. A natural extension is to investigate whether the oracle-efficient distribution for the high-dimensional adaptive Lasso in the cross-sectional setting ([Huang et al., 2008b](#)) can be generalized to the linear panel data framework with random regressors, under large N and fixed T , with or without restrictions on parameter dimensionality.
2. The idea of weighted regularization can be extended to nonlinear panel models. For instance, future research may explore whether weighted regularization improves the asymptotic properties of regularized quantile fixed effects estimator as studied in [Koenker \(2004\)](#) and [Galvao and Montes-Rojas \(2010\)](#). Given the success of the adaptive Lasso in general M -estimation, it is plausible that some of its desirable properties could carry over to nonlinear settings.
3. In empirical applications, fixed effects often appear in multiple dimensions rather than solely at the individual level. Beyond the widely used two-way fixed effects models, an emerging literature addresses multi-way fixed effects structures. For example, [Ortega and Peri \(2013\)](#) study the impact of income on international migration using a model with fixed effects of the form $\alpha_i + \gamma_j + \delta_t + \tau_{it} + \iota_{ij}$, where i and j index countries and t indexes years. In such settings, identification, asymptotics, and the incidental parameters problem become considerably more complex. We believe that weighted regularization holds promise for improving both theoretical properties and practical performance of multi-way fixed effects estimators.
4. Finally, when individual fixed effects exhibit latent group structure—as in the grouped fixed effects model of [Bonhomme and Manresa \(2015\)](#)—regularization techniques may offer an alternative solution to simultaneously estimate the parameters of interest and uncover the underlying group structure. This dual role of regularization presents an exciting direction for future methodological development.

A Proofs and Lemmas

Proof of Theorem 1. By the definition of $\hat{\gamma}$, we have

$$\frac{1}{n} \|\mathbf{y} - \mathbf{C}\hat{\gamma}\|_2^2 + \lambda_n \sum_{i=1}^N w_i |\hat{\alpha}_i| \leq \frac{1}{n} \|\mathbf{y} - \mathbf{C}\gamma^*\|_2^2 + \lambda_n \sum_{i \in s} w_i |\alpha_i^*|. \quad (27)$$

By rewrite the inequality, we have

$$\frac{1}{n} \|\mathbf{C}(\hat{\gamma} - \gamma^*)\|_2^2 + \lambda_n \sum_{i=1}^N w_i |\alpha_i^*| \leq \frac{2}{n} \varepsilon' \mathbf{C}(\hat{\gamma} - \gamma^*) + \lambda_n \sum_{i \in s} w_i |\alpha_i^*| \quad (28)$$

$$\leq M_0 \|\hat{\gamma} - \gamma^*\|_1 + \lambda_n \sum_{i \in s} w_i |\alpha_i^*| \quad (29)$$

$$= M_0 \|\hat{\gamma}_s - \gamma^*\|_1 + M_0 \|\hat{\alpha}_z\|_1 + \lambda_n \sum_{i \in s} w_i |\alpha_i^*|. \quad (30)$$

For the second term in the LHS of the above inequality, note that

$$\lambda_n \sum_{i=1}^N w_i |\hat{\alpha}_i| = \lambda_n \sum_{i \in s} w_i |\hat{\alpha}_i| + 2 \left(\frac{\lambda_n}{2} \sum_{i \in z} w_i |\hat{\alpha}_i| \right) \quad (31)$$

$$\geq \lambda_n \sum_{i \in s} w_i |\hat{\alpha}_i| + \frac{\lambda_n}{2} \sum_{i \in z} w_i |\hat{\alpha}_i| + \frac{\lambda_n}{2} w_z^{\min} \|\hat{\alpha}_z\|_1 \quad (32)$$

$$\geq \lambda_n \sum_{i \in s} w_i |\hat{\alpha}_i| + \frac{\lambda_n}{2} \sum_{i \in z} w_i |\hat{\alpha}_i| + M_0 \|\hat{\alpha}_z\|_1. \quad (33)$$

Therefore,

$$\begin{aligned} \frac{1}{n} \|\mathbf{C}(\hat{\gamma} - \gamma^*)\|_2^2 + \frac{\lambda_n}{2} \sum_{i \in z} w_i |\hat{\alpha}_i| &\leq M_0 \|\hat{\gamma}_s - \gamma^*\|_1 + \lambda_n \sum_{i \in s} w_i (|\alpha_i^*| - |\hat{\alpha}_i|) \\ &\leq M_0 \sqrt{p + s_N} \|\hat{\gamma}_s - \gamma^*\|_2 + \lambda_n \sum_{i \in s} w_i |\hat{\alpha}_i - \alpha_i^*| \\ &\leq \frac{\lambda_n}{2} \|\mathbf{w}_s\|_2 \|\hat{\gamma}_s - \gamma^*\|_2 + \lambda_n \|\mathbf{w}_s\|_2 \|\hat{\alpha}_s - \alpha^*\|_2 \\ &\leq \frac{3}{2} \lambda_n \|\mathbf{w}_s\|_2 \|\hat{\gamma}_s - \gamma^*\|_2. \end{aligned}$$

The above inequality also implies

$$\frac{\lambda_n}{2} w_z^{\min} \|\hat{\alpha}_z\|_1 \leq \frac{\lambda_n}{2} \sum_{i \in z} w_i |\hat{\alpha}_i| \leq \frac{3}{2} \lambda_n \sqrt{s_N} w_z^{\min} \|\hat{\gamma}_s - \gamma^*\|_2, \quad (34)$$

so that

$$\|(\hat{\gamma} - \gamma^*)_{s^c}\|_1 \leq 3\sqrt{s_N} \|(\hat{\gamma} - \gamma^*)_s\|_2. \quad (35)$$

By the restricted eigenvalue condition, we have

$$\begin{aligned}
\frac{1}{n} \|\mathbf{C}(\hat{\gamma} - \gamma^*)\|_2^2 + \frac{\lambda_n}{2} \sum_{i \in z} w_i |\hat{\alpha}_i| + \frac{1}{2} \lambda_n \|\mathbf{w}_s\|_2 \|\hat{\gamma}_s - \gamma^*\|_2 &\leq 2\lambda_n \|\mathbf{w}_s\|_2 \|\hat{\gamma}_s - \gamma^*\|_2 \\
&\leq \frac{2\lambda_n \|\mathbf{w}_s\|_2}{\phi} \frac{\|\mathbf{C}(\hat{\gamma} - \gamma^*)\|_2}{\sqrt{n}} \\
&\leq \frac{\|\mathbf{C}(\hat{\gamma} - \gamma^*)\|_2^2}{n} + \frac{\lambda_n^2 \|\mathbf{w}_s\|_2^2}{\phi^2}.
\end{aligned}$$

Lemma 1. Let $\|\cdot\|_{\psi_2}$ denote the Orlicz norm associated with the function $\psi_2(x) = e^{x^2} - 1$. If $\sum_{i=1}^N \sum_{t=1}^T a_{it}^2 \leq M_1$, then

$$\left\| \sum_{i=1}^N \sum_{t=1}^T a_{it} \varepsilon_{it} \right\|_{\psi_2} \leq M_2, \quad (36)$$

where M_2 depends on A, B, M_1 , and σ .

Proof. The tail bound on ε_{it} implies

$$\|\varepsilon_{it}\|_{\psi_2} \leq \left(\frac{1+A}{B} \right)^{1/2}.$$

By the proposition 1.6 in [Van der Vaart and Wellner \(2023\)](#), there exists some $M_3 > 0$ such that

$$\left\| \sum_{i=1}^N \sum_{t=1}^T a_{it} \varepsilon_{it} \right\|_{\psi_2} \leq M_3 \left(\mathbb{E} \left| \sum_{i=1}^N \sum_{t=1}^T a_{it} \varepsilon_{it} \right| + \left(\sum_{i=1}^N \sum_{t=1}^T \|a_{it} \varepsilon_{it}\|_{\psi_2}^2 \right)^{1/2} \right) \quad (37)$$

$$\leq M_3 \left(M_1^{1/2} \left(\sigma + \left(\frac{1+A}{B} \right)^{1/2} \right) \right). \quad (38)$$

□

Lemma 2. Let $\{V_j\}_{j=1}^k$ be random variables such that $\|V_j\|_{\psi_2} \leq M_4$ for all $1 \leq j \leq k$. Then, for all $v > 0$, the following tail bound holds:

$$\mathbb{P} \left(\max_{1 \leq j \leq k} |V_j| \geq v \right) \lesssim \frac{\sqrt{\log k}}{v}. \quad (39)$$

Proof. Note that $\mathbb{E}|V_j| \leq \sqrt{\log 2} \|V_j\|_{\psi_2}$ in general. Therefore,

$$\mathbb{E} \left[\max_{1 \leq j \leq k} |V_j| \right] \leq \sqrt{\log 2} \left\| \max_{1 \leq j \leq k} |V_j| \right\|_{\psi_2} \quad (40)$$

$$\leq \sqrt{2} M_5 \sqrt{\log k} \max_{1 \leq j \leq k} \|V_j\|_{\psi_2} \quad (41)$$

$$\leq \sqrt{2} M_4 M_5 \sqrt{\log k}, \quad (42)$$

where the second inequality follows from the property of the Orlicz norm (see section 2.2 in [Van der Vaart and Wellner \(2023\)](#)), and M_5 is some universal constant. Then the upper bound follows from the Markov's inequality. □

Lemma 3. Recall that $\mathbf{R}_X = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, define $\tilde{\mathbf{y}} := \mathbf{R}_X \mathbf{y}$, $\tilde{\mathbf{D}}_s := \mathbf{R}_X \mathbf{D}_s$, $\tilde{\mathbf{D}}_z := \mathbf{R}_X \mathbf{D}_z$,

$\tilde{\Sigma}_{11} := \frac{1}{n} \tilde{\mathbf{D}}'_s \tilde{\mathbf{D}}_s$, $\tilde{\Sigma}_{21} := \frac{1}{n} \tilde{\mathbf{D}}'_z \tilde{\mathbf{D}}_s$, and $\mathbf{W}_s := \text{diag}(\mathbf{w}_s)$. Define the events,

$$\mathcal{A}_n := \left\{ \frac{2}{\sqrt{n}} \left| \tilde{\Sigma}_{11}^{-1} \tilde{\mathbf{D}}'_s \boldsymbol{\varepsilon} \right| < 2\sqrt{n} |\boldsymbol{\alpha}_s^*| - \frac{\lambda_n}{\sqrt{n}} \left| \tilde{\Sigma}_{11}^{-1} \mathbf{W}_s \text{sgn}(\boldsymbol{\alpha}_s^*) \right| \right\}, \quad (43)$$

$$\mathcal{B}_n = \left\{ \frac{2}{\sqrt{n}} \left| \tilde{\mathbf{D}}'_z \mathbf{R}_{\tilde{\mathbf{D}}_s} \boldsymbol{\varepsilon} \right| \leq \frac{\lambda_n}{\sqrt{n}} \mathbf{w}_z - \frac{\lambda_n}{\sqrt{n}} \left| \tilde{\Sigma}_{21} \tilde{\Sigma}_{11}^{-1} \mathbf{W}_s \text{sgn}(\boldsymbol{\alpha}_s^*) \right| \right\}. \quad (44)$$

The probability of $\text{sgn}(\hat{\boldsymbol{\alpha}}) = \text{sgn}(\boldsymbol{\alpha}^*)$ is bounded by the probability of intersection of $\mathcal{A}_n$ and $\mathcal{B}_n$, i.e.,

$$\mathbb{P}(\text{sgn}(\hat{\boldsymbol{\alpha}}) = \text{sgn}(\boldsymbol{\alpha}^*)) \geq \mathbb{P}(\mathcal{A}_n \cap \mathcal{B}_n). \quad (45)$$

Proof. This proof is based on a modification of the proof of Proposition 1 in [Zhao and Yu \(2006\)](#).

We first note that given $\hat{\boldsymbol{\beta}}$ is known, we can write

$$\hat{\boldsymbol{\alpha}} = \underset{\boldsymbol{\alpha} \in \mathbb{R}^N}{\text{argmin}} \left\| \mathbf{R}_X \mathbf{y} - \mathbf{R}_X \mathbf{D} \boldsymbol{\alpha} \right\|_2^2 + \lambda_n \|\mathbf{W} \boldsymbol{\alpha}\|_1. \quad (46)$$

Let

$$\hat{\boldsymbol{\alpha}}_s := (\tilde{\mathbf{D}}'_s \tilde{\mathbf{D}}_s)^{-1} \left(\tilde{\mathbf{D}}'_s \tilde{\mathbf{y}} - \frac{\lambda_n}{2} \mathbf{W}_s \text{sgn}(\boldsymbol{\alpha}_s^*) \right) = \boldsymbol{\alpha}_s^* + \frac{1}{\sqrt{n}} \tilde{\Sigma}_{11}^{-1} \left(\frac{\tilde{\mathbf{D}}'_s \boldsymbol{\varepsilon}}{\sqrt{n}} + \frac{\lambda_n}{2\sqrt{n}} \mathbf{W}_s \text{sgn}(\boldsymbol{\alpha}_s^*) \right), \quad (47)$$

and $\hat{\boldsymbol{\alpha}} := (\hat{\boldsymbol{\alpha}}'_s, \mathbf{0}')'$.

First note that inequality in $\mathcal{A}_n$ implies that

$$\left| \boldsymbol{\alpha}_s^* - \left[\boldsymbol{\alpha}_s^* + \frac{1}{\sqrt{n}} \tilde{\Sigma}_{11}^{-1} \left(\frac{\tilde{\mathbf{D}}'_s \boldsymbol{\varepsilon}}{\sqrt{n}} + \frac{\lambda_n}{2\sqrt{n}} \mathbf{W}_s \text{sgn}(\boldsymbol{\alpha}_s^*) \right) \right] \right| < \boldsymbol{\alpha}_s^*. \quad (48)$$

Substitute the definition of $\hat{\boldsymbol{\alpha}}_s$, we have $|\boldsymbol{\alpha}_s^* - \hat{\boldsymbol{\alpha}}_s| < |\boldsymbol{\alpha}_s^*|$, which holds only if $\text{sgn}(\hat{\boldsymbol{\alpha}}_s) = \text{sgn}(\boldsymbol{\alpha}_s^*)$.

On the other hand, the inequality in $\mathcal{B}_n$ implies that

$$\left| \tilde{\mathbf{D}}'_z \mathbf{R}_{\tilde{\mathbf{D}}_s} \boldsymbol{\varepsilon} + \frac{\lambda_n}{2} \tilde{\Sigma}_{21} \tilde{\Sigma}_{11}^{-1} \mathbf{W}_s \text{sgn}(\boldsymbol{\alpha}_s^*) \right| < \frac{\lambda_n}{2} \mathbf{w}_z. \quad (49)$$

By rewrite the LHD, we have

$$\tilde{\mathbf{D}}'_z \left(\tilde{\mathbf{y}} - \tilde{\mathbf{D}}_z \left[(\tilde{\mathbf{D}}'_s \tilde{\mathbf{D}}_s)^{-1} \left(\tilde{\mathbf{D}}'_s \tilde{\mathbf{y}} - \frac{\lambda_n}{2} \mathbf{W}_s \text{sgn}(\boldsymbol{\alpha}_s^*) \right) \right] \right) \leq \frac{\lambda_n}{2} \mathbf{w}_z. \quad (50)$$

Put everything together, and by our definition of $\hat{\boldsymbol{\alpha}}$, we see that conditions in $\mathcal{A}_n$ and $\mathcal{B}_n$ imply that $\hat{\boldsymbol{\alpha}}$ satisfies the conditions

$$\tilde{\mathbf{D}}'_s (\tilde{\mathbf{y}} - \tilde{\mathbf{D}}_s \hat{\boldsymbol{\alpha}}_s) = \frac{\lambda_n}{2} \mathbf{W}_s \text{sgn}(\hat{\boldsymbol{\alpha}}_s) \quad \text{and} \quad \left| \tilde{\mathbf{D}}'_z (\tilde{\mathbf{y}} - \tilde{\mathbf{D}}_s \hat{\boldsymbol{\alpha}}_s) \right| \leq \frac{\lambda_n}{2} \mathbf{w}_z. \quad (51)$$

Note that conditions (45) is the Karush-Kuhn-Tucker (KKT) conditions of the minimization defined in (40). By the uniqueness of the minimizer of (40), we see that $\hat{\boldsymbol{\alpha}}$ satisfies the property $\text{sgn}(\hat{\boldsymbol{\alpha}}) = \text{sgn}(\boldsymbol{\alpha}^*)$. $\square$

Lemma 4. Let

$$\boldsymbol{\eta} = (\eta_1, \dots, \eta_{s_N})' := \frac{2}{\sqrt{n}} \tilde{\Sigma}_{11}^{-1} \tilde{\mathbf{D}}'_s \boldsymbol{\varepsilon} \quad \text{and} \quad \boldsymbol{\nu} = (\nu_1, \dots, \nu_{z_N})' := \frac{2}{\sqrt{n}} \tilde{\mathbf{D}}'_z \mathbf{R}_{\tilde{\mathbf{D}}_s} \boldsymbol{\varepsilon}. \quad (52)$$

For $j = 1, \dots, s_N$, we have $\eta_j = \sum_{i=1}^N \sum_{t=1}^T \eta_{jit} \varepsilon_{it}$ and $\sum_{i=1}^N \sum_{t=1}^T \eta_{jit}^2 \leq M_6$ for some constant M_6 . For $l = 1, \dots, z_N$, we have $\nu_l = \sum_{i=1}^N \sum_{t=1}^T \nu_{lit} \varepsilon_{it}$ and $\sum_{i=1}^N \sum_{t=1}^T \nu_{lit}^2 \leq M_7$ for some constant M_7 .

Proof. We let e_j and e_l to denote the j th and l th standard basis in $\mathbb{R}^{s_N}$ and $\mathbb{R}^{z_N}$ respectively. Note that

$$\sum_{i=1}^N \sum_{t=1}^T \eta_{jit}^2 = \frac{4}{n} e_j' \tilde{\Sigma}_{11}^{-1} \tilde{\mathbf{D}}_s' \tilde{\mathbf{D}}_s \tilde{\Sigma}_{11}^{-1} e_j = 4 e_j' \tilde{\Sigma}_{11}^{-1} e_j \leq 4 (\Lambda^{\min}(\frac{1}{n} \mathbf{D}_s' \mathbf{R}_s \mathbf{D}_s))^{-1} \leq \frac{4}{c^{\min}}, \quad (53)$$

$$\sum_{i=1}^N \sum_{t=1}^T \nu_{lit}^2 = \frac{4}{n} e_l' \tilde{\mathbf{D}}_z' \mathbf{R}_{\tilde{\mathbf{D}}_s} \tilde{\mathbf{D}}_z e_l \leq \frac{4}{n} e_l' \tilde{\mathbf{D}}_z' \tilde{\mathbf{D}}_z e_l \leq \frac{4}{n} e_l' \mathbf{D}_z' \mathbf{D}_z e_l = \frac{4T}{n} \leq 4. \quad (54)$$

□

Proof of Theorem 2. Follow the notation in lemma 3, we first write

$$\mathbb{P}(\mathcal{A}_n^c) = \mathbb{P}(\mathcal{A}_n^c \cap \{\mathbf{w}_z \leq 1/a_N^{\min}\}) + \mathbb{P}(\mathcal{A}_n^c \cap \{\mathbf{w}_z > 1/a_N^{\min}\}). \quad (55)$$

The second term on RHS converges to zero by construction. We show that the first term on RHS converges to zero. Note that for $1 \leq j \leq s_N$,

$$\left[\left| \tilde{\Sigma}_{11}^{-1} \mathbf{W}_s \operatorname{sgn}(\boldsymbol{\alpha}_s^*) \right| \right]_j \leq \left\| \tilde{\Sigma}_{11}^{-1} \mathbf{W}_s \operatorname{sgn}(\boldsymbol{\alpha}_s^*) \right\|_2 \quad (56)$$

$$\leq \left\| \tilde{\Sigma}_{11}^{-1} \right\|_2 \left\| \mathbf{W}_s \right\|_2 \left\| \operatorname{sgn}(\boldsymbol{\alpha}_s^*) \right\|_2 \quad (57)$$

$$\leq \frac{\sqrt{s_N}}{\Lambda^{\min}(\tilde{\Sigma}_{11}) a_N^{\min}} \quad (58)$$

$$\leq \frac{\sqrt{s_N}}{c^{\min} a_N^{\min}}. \quad (59)$$

Therefore,

$$\begin{aligned} \mathcal{A}_n^c \cap \{\mathbf{w}_s \leq 1/a_N^{\min}\} &\subseteq \left\{ \frac{2}{\sqrt{n}} \left| \tilde{\Sigma}_{11}^{-1} \tilde{\mathbf{D}}_s' \boldsymbol{\varepsilon} \right| \geq 2\sqrt{n} a_N^{\min} - \frac{\lambda_n}{\sqrt{n}} \frac{\sqrt{s_N}}{c^{\min} a_N^{\min}} \right\} \\ &\subseteq \left\{ \max_{1 \leq j \leq s_N} |\eta_j| \geq 2\sqrt{n} a_N^{\min} - \frac{1}{c^{\min}} \frac{\lambda_n \sqrt{s_N}}{\sqrt{n} a_N^{\min}} \right\}. \end{aligned}$$

By lemma 4, condition for lemma 1 is satisfied, so that $\|\eta_j\|_{\psi_2} \leq M_2$ for $1 \leq j \leq s_N$. By lemma 2,

$$\begin{aligned} \mathbb{P}(\mathcal{A}_n^c \cap \{\mathbf{w}_s \leq 1/a_N^{\min}\}) &\lesssim \frac{\sqrt{\log s_N}}{2\sqrt{n} a_N^{\min} - \frac{1}{c^{\min}} \frac{\lambda_n \sqrt{s_N}}{\sqrt{n} a_N^{\min}}} \\ &= \frac{\sqrt{\log s_N}}{\sqrt{NT} a_N^{\min}} \frac{1}{2 - \frac{\lambda_n \sqrt{s_N}}{NT(a_N^{\min})^2}} \rightarrow 0. \end{aligned}$$

Similarly, we write

$$\mathbb{P}(\mathcal{B}_n^c) = \mathbb{P}(\mathcal{B}_n^c \cap \{\mathbf{w}_z \geq r_n \text{ and } \mathbf{w}_s \leq 1/a_N^{\min}\}) + \mathbb{P}(\mathcal{B}_n^c \cap \{\mathbf{w}_z < r_n \text{ or } \mathbf{w}_s > 1/a_N^{\min}\}).$$

The second term on LHS converges to zero by construction. We show that the first term on RHS

converges to zero. Note that for $1 \leq l \leq z_N$,

$$\begin{aligned}
\left| \left[\tilde{\Sigma}_{21} \tilde{\Sigma}_{11}^{-1} \mathbf{W}_s \operatorname{sgn}(\boldsymbol{\alpha}_s^*) \right]_l \right| &= \sum_{j=1}^{s_N} \frac{[\mathbf{R}_x \mathbf{D}_z]_l' [\mathbf{R}_x \mathbf{D}_s]_j}{n} \left[\tilde{\Sigma}_{11}^{-1} \mathbf{W}_s \operatorname{sgn}(\boldsymbol{\alpha}_s^*) \right]_j \\
&\leq \sum_{j=1}^{s_N} \frac{\|[\mathbf{R}_x \mathbf{D}_z]_l\|_2 \|[\mathbf{R}_x \mathbf{D}_s]_j\|_2}{n} \frac{\sqrt{s_N}}{c^{\min} a_N^{\min}} \\
&\leq \sum_{j=1}^{s_N} \frac{T}{NT} \frac{\sqrt{s_N}}{c^{\min} a_N^{\min}} \\
&\leq \frac{1}{c^{\min}} \frac{s_N^2}{N a_N^{\min}}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\mathbb{P}(\mathcal{B}_n^c \cap \{\mathbf{w}_z \geq r_n \text{ and } \mathbf{w}_s \leq 1/a_N^{\min}\}) &\subseteq \left\{ \frac{2}{\sqrt{n}} \left| \tilde{\mathbf{D}}_z' \mathbf{R}_{\tilde{\mathbf{D}}_s} \boldsymbol{\varepsilon} \right| > \frac{\lambda_n}{\sqrt{n}} r_n - \frac{\lambda_n}{\sqrt{n}} \frac{s_N^2}{c^{\min} N a_N^{\min}} \right\} \\
&\subseteq \left\{ \max_{1 \leq l \leq z_N} |\nu_l| > \frac{\lambda_n r_n}{\sqrt{n}} - \frac{\lambda_n}{\sqrt{n}} \frac{s_N^2}{c^{\min} N a_N^{\min}} \right\}.
\end{aligned}$$

Again by lemma 4, condition for lemma 1 is satisfied, so that $\|\nu_l\|_{\psi_2} \leq M_2$ for $1 \leq l \leq z_N$. By lemma 2

$$\begin{aligned}
\mathbb{P}(\mathcal{B}_n^c \cap \{\mathbf{w}_z \geq r_n \text{ and } \mathbf{w}_s \leq 1/a_N^{\min}\}) &\lesssim \frac{\sqrt{\log z_N}}{\frac{\lambda_n r_n}{\sqrt{n}} - \frac{\lambda_n}{\sqrt{n}} \frac{s_N^2}{c^{\min} N a_N^{\min}}} \\
&= \frac{\sqrt{NT \log z_N}}{\lambda_n r_n} \frac{1}{1 - \frac{1}{c^{\min}} \frac{s_N^2}{r_n N a_N^{\min}}} \rightarrow 0.
\end{aligned}$$

By lemma 3, theorem 2 follows from the fact that

$$\mathbb{P}(\operatorname{sgn}(\hat{\boldsymbol{\alpha}}) = \operatorname{sgn}(\boldsymbol{\alpha}^*)) \geq \mathbb{P}(\mathcal{A}_n \cap \mathcal{B}_n) = 1 - \mathbb{P}(\mathcal{A}_n^c) - \mathbb{P}(\mathcal{B}_n^c) \rightarrow 1. \quad (60)$$

Proof of Theorem 3. We note that when $\operatorname{sgn}(\hat{\boldsymbol{\alpha}}) = \operatorname{sgn}(\boldsymbol{\alpha}^*)$, the solution of (4) can be characterized by the KKT conditions:

$$\mathbf{X}'(\mathbf{y} - \mathbf{C}_s \hat{\boldsymbol{\gamma}}_s) = 0, \quad \mathbf{D}'_x(\mathbf{y} - \mathbf{C}_s \hat{\boldsymbol{\gamma}}_s) = \frac{\lambda_n}{2} \mathbf{W}_s \operatorname{sgn}(\hat{\boldsymbol{\alpha}}_s), \quad \text{and} \quad |\mathbf{D}'_z(\mathbf{y} - \mathbf{C}_s \hat{\boldsymbol{\gamma}}_s)| \leq \frac{\lambda_n}{2} \mathbf{w}_z. \quad (61)$$

Define $\boldsymbol{\xi} := (\mathbf{0}', \mathbf{W}_s \operatorname{sgn}(\hat{\boldsymbol{\alpha}}_s)')'$, where $\mathbf{0}$ is a $p \times 1$ vector of zeros, then we could combine the first and second equation as

$$\mathbf{C}'_s(\mathbf{y} - \mathbf{C}_s \hat{\boldsymbol{\gamma}}_s) = \frac{\lambda_n}{2} \boldsymbol{\xi}. \quad (62)$$

By theorem 2, we have

$$\mathbb{P} \left(\hat{\boldsymbol{\gamma}}_s - \boldsymbol{\gamma}_s^* = (\mathbf{C}'_s \mathbf{C}_s)^{-1} \mathbf{C}'_s \boldsymbol{\varepsilon} - \frac{\lambda_n}{2} (\mathbf{C}'_s \mathbf{C}_s)^{-1} \boldsymbol{\xi} \right) \rightarrow 1. \quad (63)$$

Therefore,

$$\sqrt{NT} v_n^{-1} \mathbf{u}'(\hat{\boldsymbol{\gamma}}_s - \boldsymbol{\gamma}_s^*) = \sum_{i=1}^N \sum_{t=1}^T \frac{1}{\sqrt{NT}} v_n^{-1} \mathbf{u}' \Sigma_{11}^{-1} \mathbf{c}_{sit} \varepsilon_{it} - \frac{\lambda_n}{2\sqrt{n}} v_n^{-1} \mathbf{u}' \Sigma_{11}^{-1} \boldsymbol{\xi} + o_p(1). \quad (64)$$

Since

$$\left| \frac{\lambda_n}{2\sqrt{n}} v_n^{-1} \mathbf{u}' \Sigma_{11}^{-1} \xi \right| \leq \frac{\lambda_n}{2\sqrt{n}} \frac{\Lambda^{\max}(\Sigma_{11}^{-1})}{\sigma \sqrt{\Lambda^{\min}(\Sigma_{11}^{-1})}} \frac{\sqrt{s_N}}{a_N^{\min}} = \frac{1}{2\sigma} \frac{\Lambda^{\max}(\Sigma_{11})}{\Lambda^{\min}(\Sigma_{11})} \frac{\lambda_n \sqrt{s_N}}{\sqrt{n} a_N^{\min}} \rightarrow 0, \quad (65)$$

it suffices to show that

$$\sum_{i=1}^N \sum_{t=1}^T \frac{1}{\sqrt{NT}} v_n^{-1} \mathbf{u}' \Sigma_{11}^{-1} \mathbf{c}_{sit} \varepsilon_{it} \xrightarrow{d} \mathcal{N}(0, 1). \quad (66)$$

This can be done by verifying a version of the Lindeberg–Feller condition, as stated in Section 2.8 of [Van der Vaart \(2000\)](#), namely,

$$\max_{i,t} \left| \frac{1}{\sqrt{NT}} v_n^{-1} \mathbf{u}' \Sigma_{11}^{-1} \mathbf{c}_{sit} \right| \rightarrow 0. \quad (67)$$

Under the assumption **B5**, this follows from

$$\max_{it} \left| \frac{1}{\sqrt{NT}} v_n^{-1} \mathbf{u}' \Sigma_{11}^{-1} \mathbf{c}_{sit} \right| \leq \frac{1}{\sigma} \frac{1}{\sqrt{NT \Lambda^{\min}(\Sigma_{11})}} \max_{it} \sqrt{\mathbf{x}_{it}' \mathbf{x}_{it} + T} \rightarrow 0. \quad (68)$$

B Additional Simulations

In this appendix, we present additional simulation results under alternative data-generating processes to assess the robustness of the Sparse Fixed Effects (SFE) estimator.

Design A: This design follows the simulation setup described in Section 4 of the main text. Here, we provide the complete set of simulation results across all parameter configurations. Summary tables are reported in the main text. Figures 5–9 present the density plots of the Oracle Ordinary Least Squares (OOLS), Fixed Effects (FE), and Sparse Fixed Effects (SFE) estimators. Figures 10–14 display the corresponding box plots for OOLS, Random Effects (RE), FE, standard Lasso (LASSO), Post-Lasso (PL), and SFE.

Design B: This design uses the same data-generating process as in Design A but with a smaller panel: $N = 50$ and $T = 5$. Figures 15–19 show the density plots of OOLS, FE, and SFE. Figures 20–24 display the box plots for OOLS, RE, FE, LASSO, PL, and SFE.

Design C: This design modifies the data-generating process in Design A by replacing $s\alpha_i^*$ with α_i^* , where

$$\alpha_i^* \sim \text{Laplace}(0, \sigma^2), \quad (69)$$

and we set $\sigma^2 = \psi$, with $\psi \in \{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$. In this setting, ψ controls the variance of the fixed effects rather than their sparsity. Figures 25–29 present the density plots of the FE and SFE estimators.

Design D: This design is identical to Design C, except the fixed effects are generated from a normal distribution:

$$\alpha_i^* \sim \mathcal{N}(0, \sigma^2). \quad (70)$$

As in Design C, $\sigma^2 = \psi$, and ψ controls the variance of the fixed effects. Figures 30–34 present the density plots of the FE and SFE estimators.

Figure 5: Design A ($\rho = 0$)

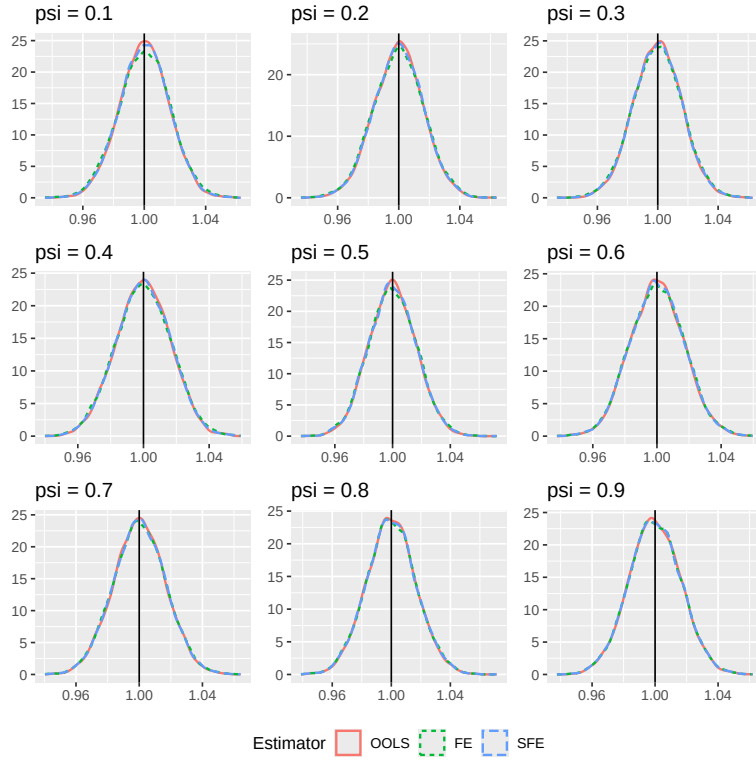


Figure 6: Design A ($\rho = 0.2$)

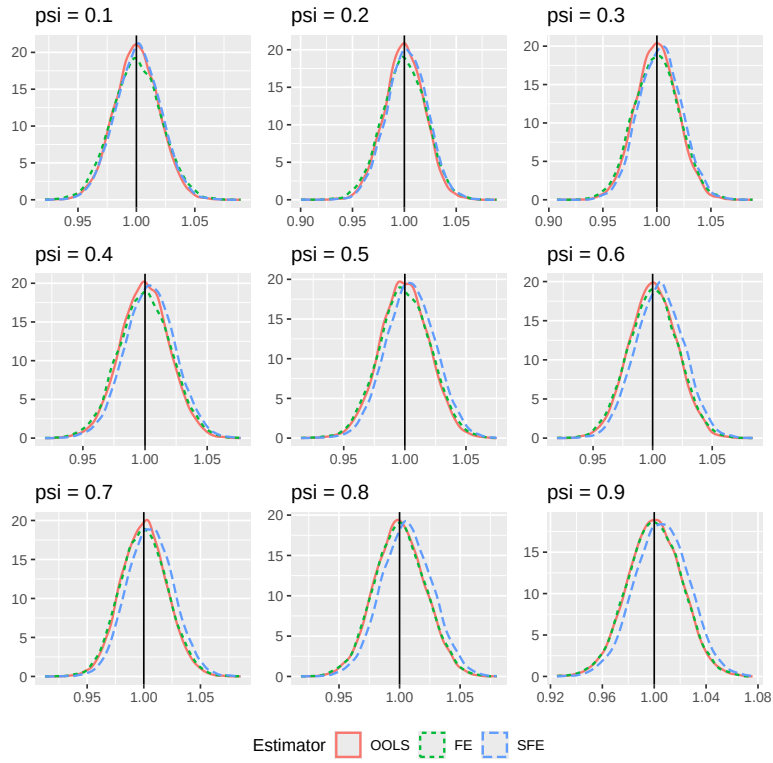


Figure 7: Design A ($\rho = 0.4$)

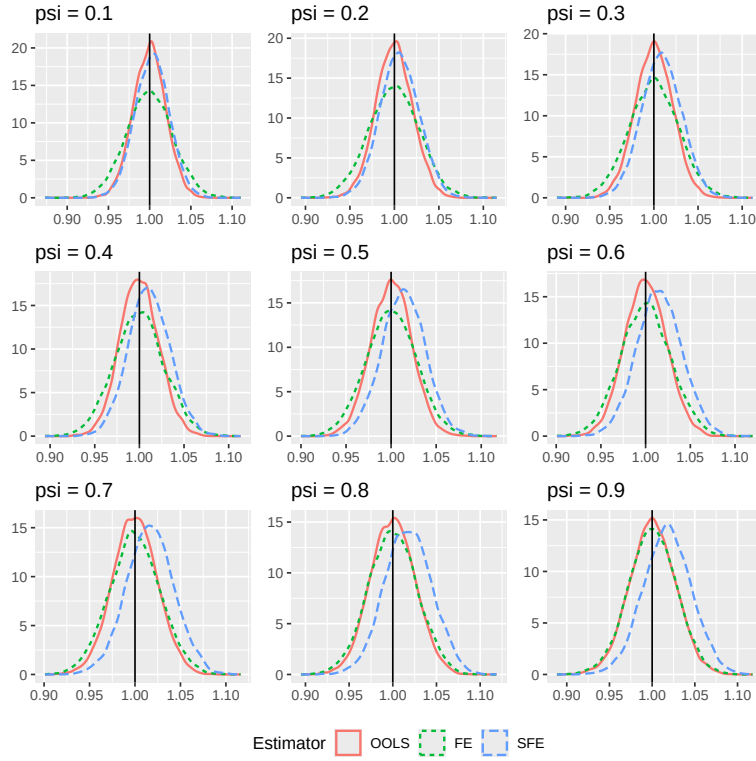


Figure 8: Design A ($\rho = 0.6$)

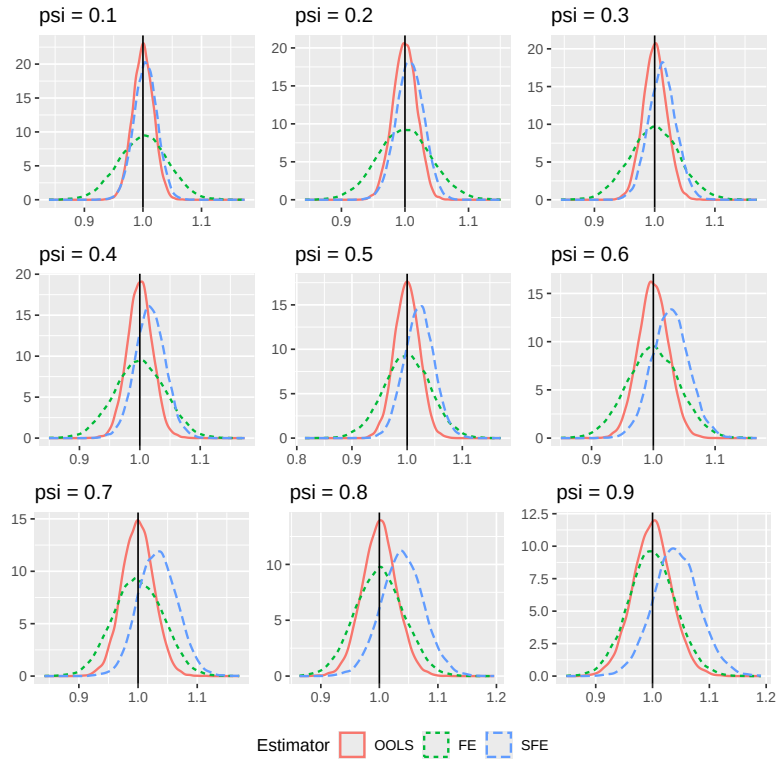


Figure 9: Design A ($\rho = 0.8$)

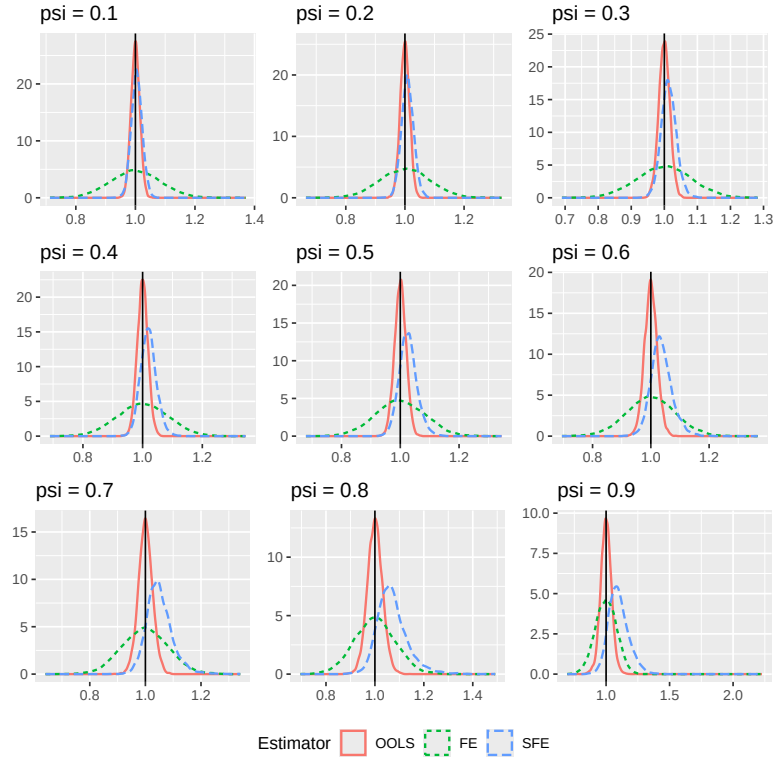


Figure 10: Design A ($\rho = 0$)

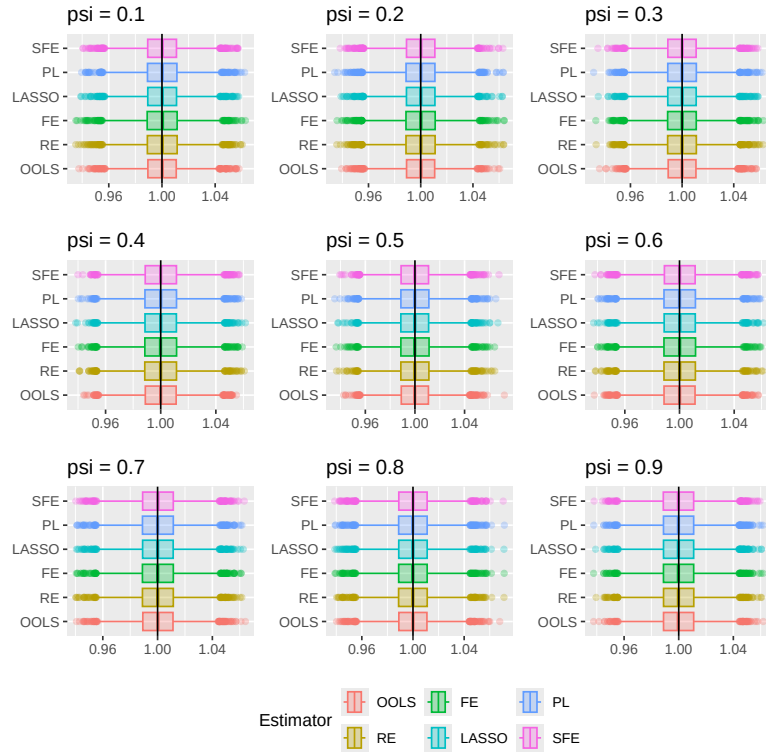


Figure 11: Design A ($\rho = 0.2$)

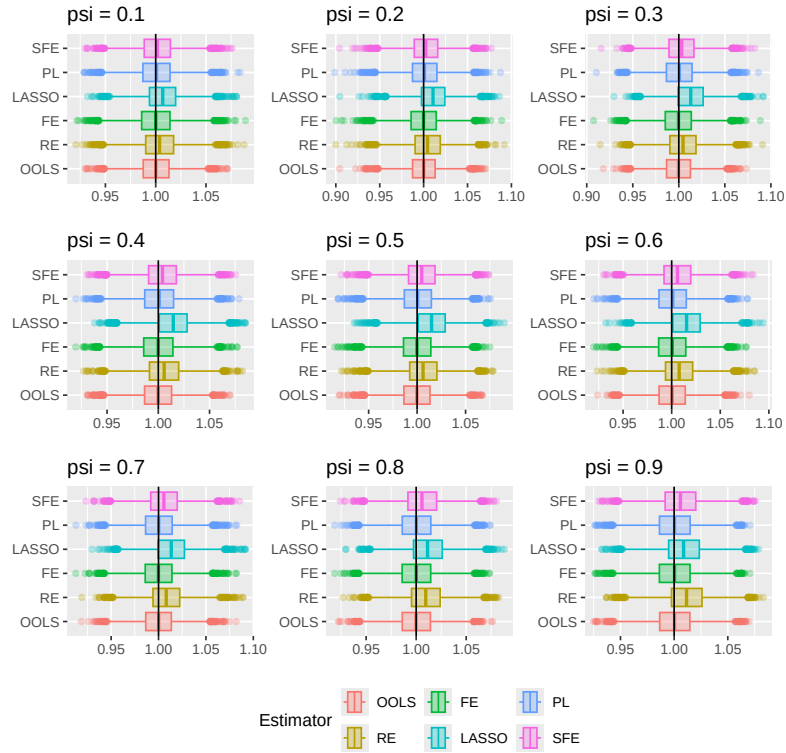


Figure 12: Design A ($\rho = 0.4$)

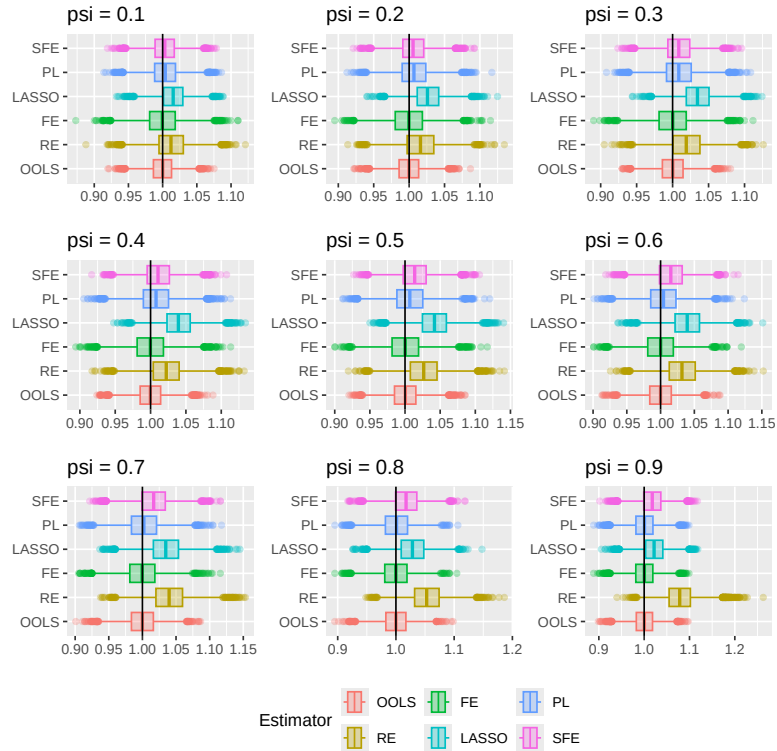


Figure 13: Design A ($\rho = 0.6$)

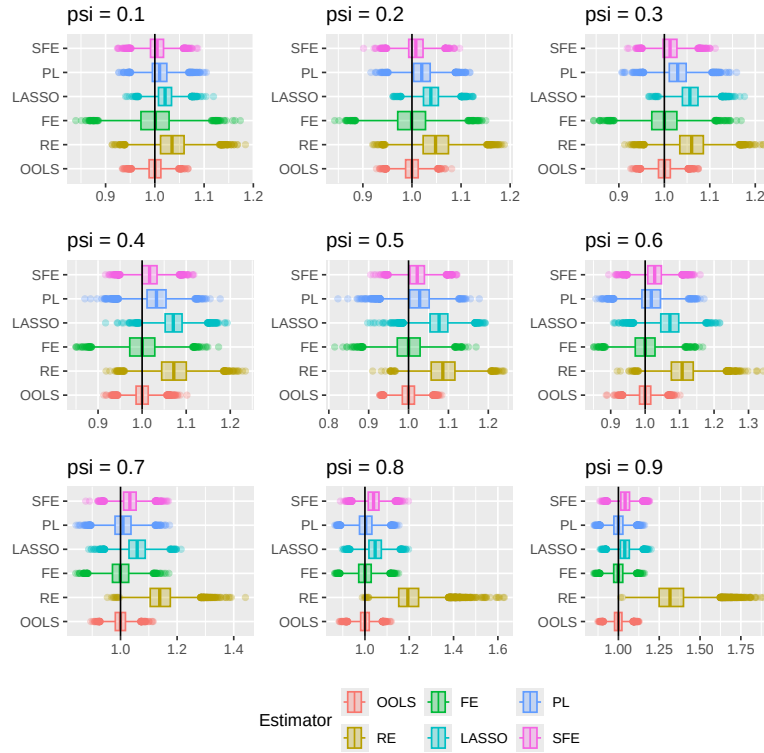


Figure 14: Design A ($\rho = 0.8$)

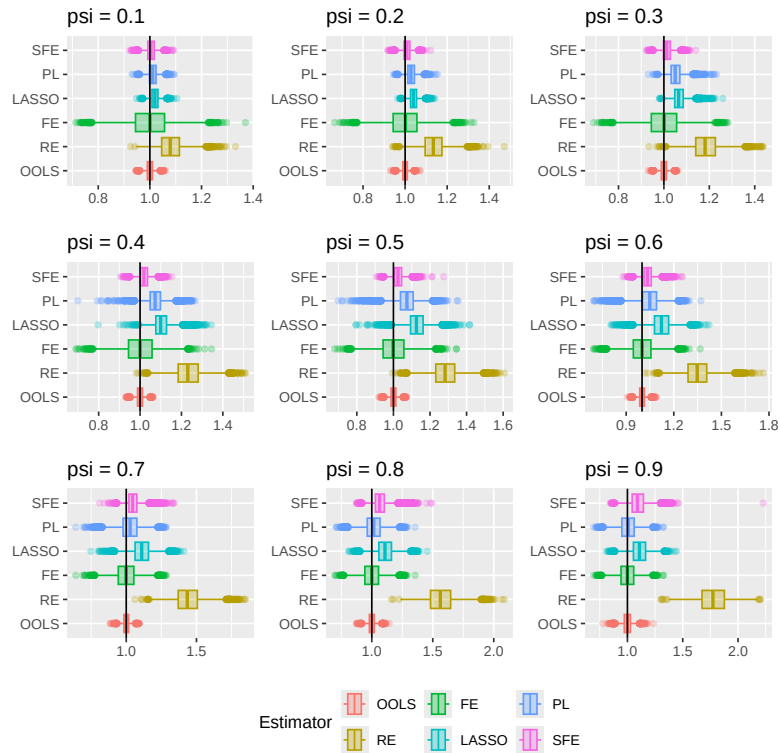


Figure 15: Design B ($\rho = 0$)

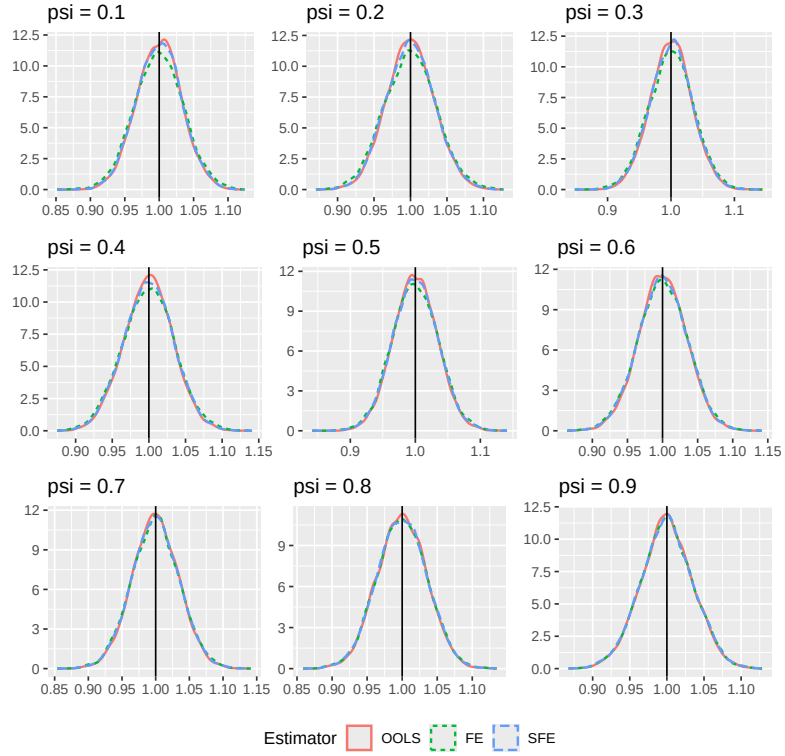


Figure 16: Design B ($\rho = 0.2$)

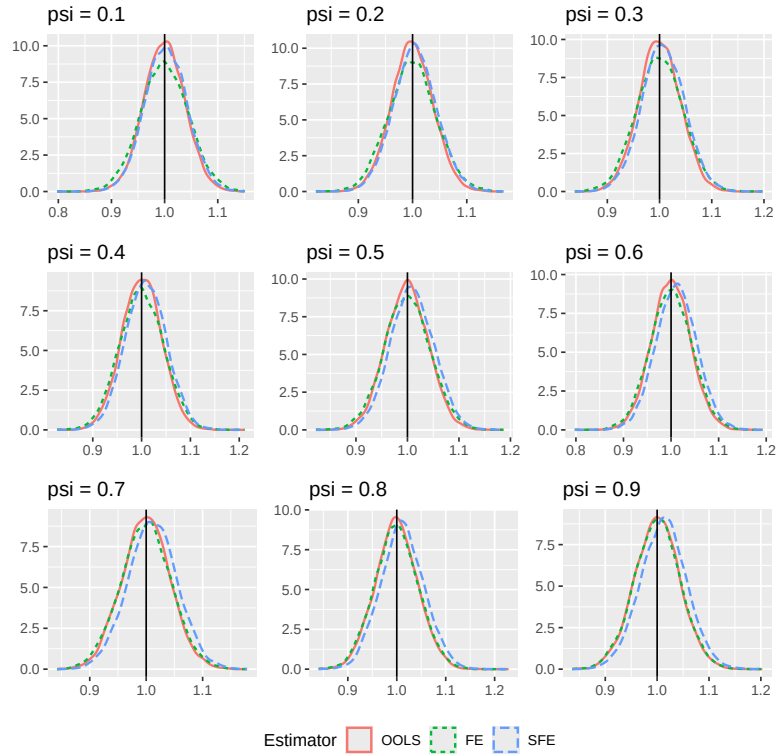


Figure 17: Design B ($\rho = 0.4$)

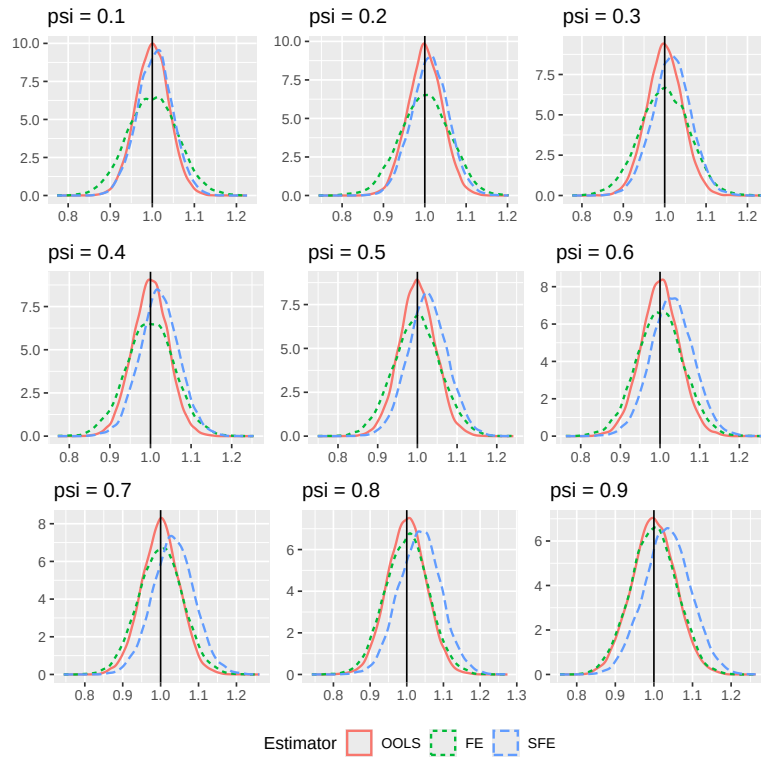


Figure 18: Design B ($\rho = 0.6$)

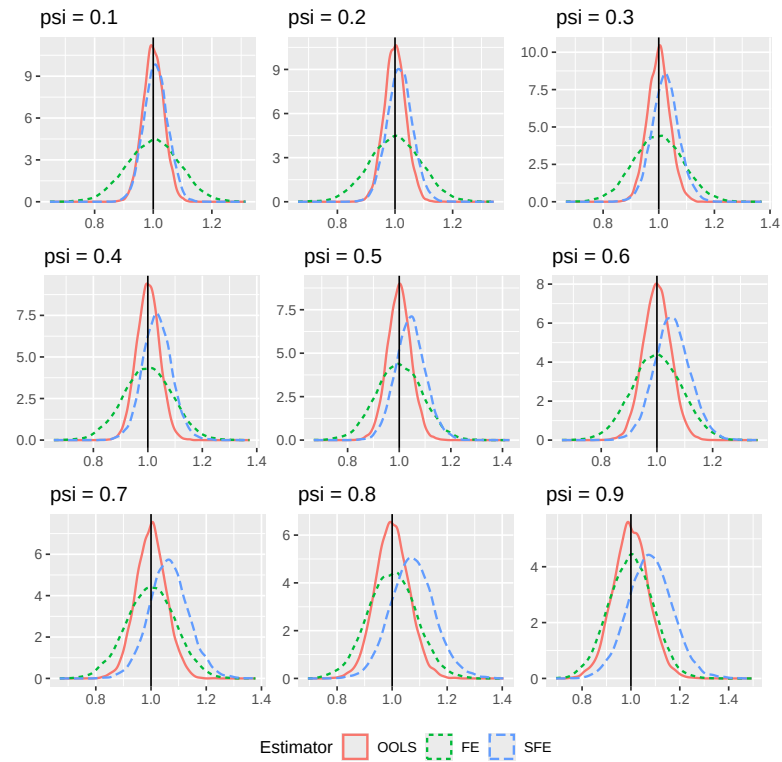


Figure 19: Design B ($\rho = 0.8$)

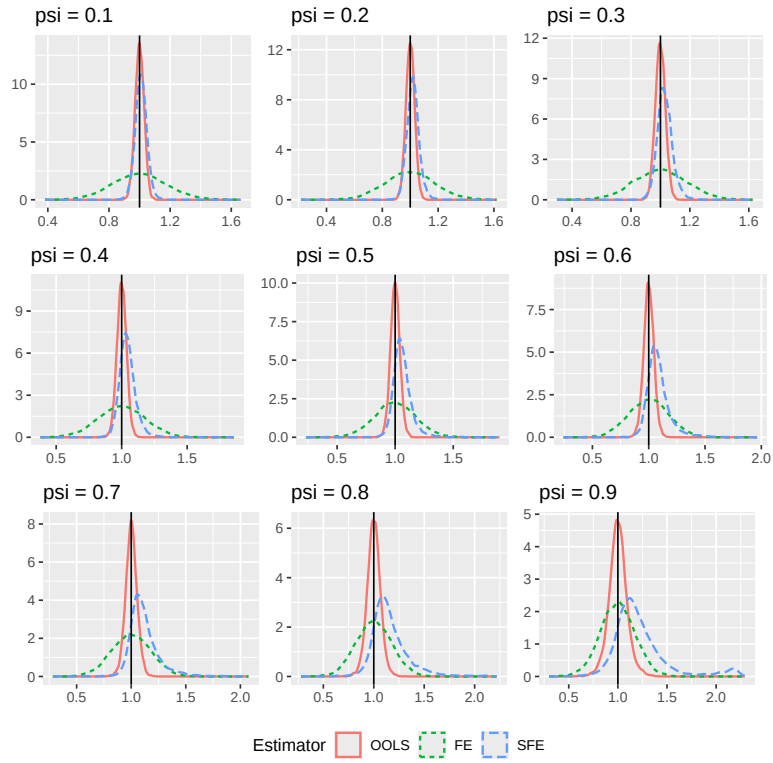


Figure 20: Design B ($\rho = 0$)

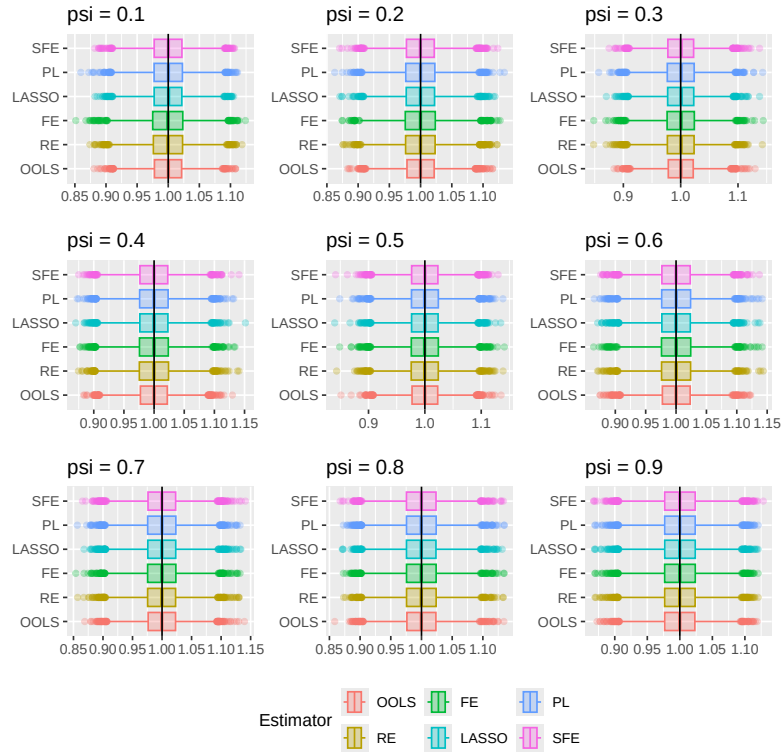


Figure 21: Design B ($\rho = 0.2$)

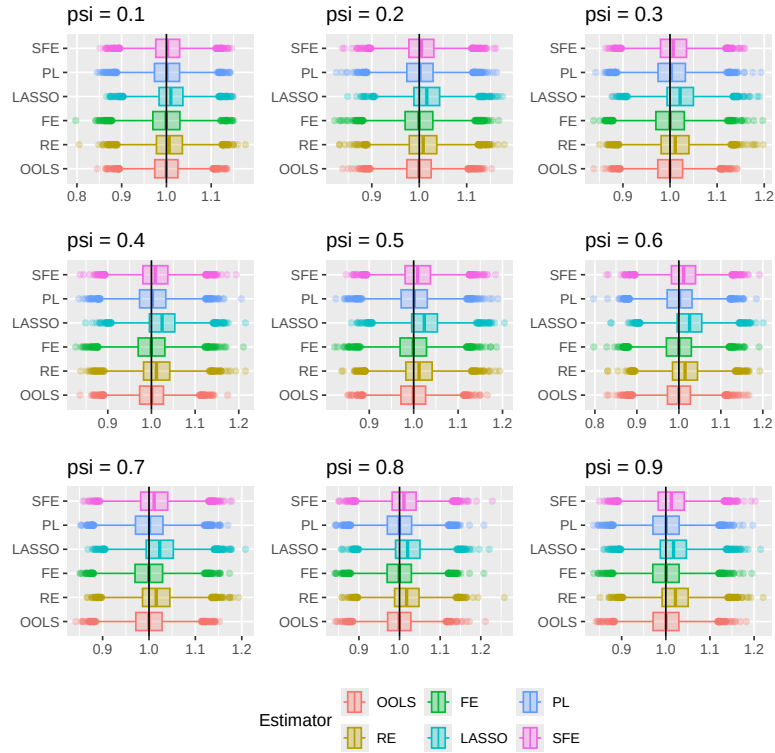


Figure 22: Design B ($\rho = 0.4$)

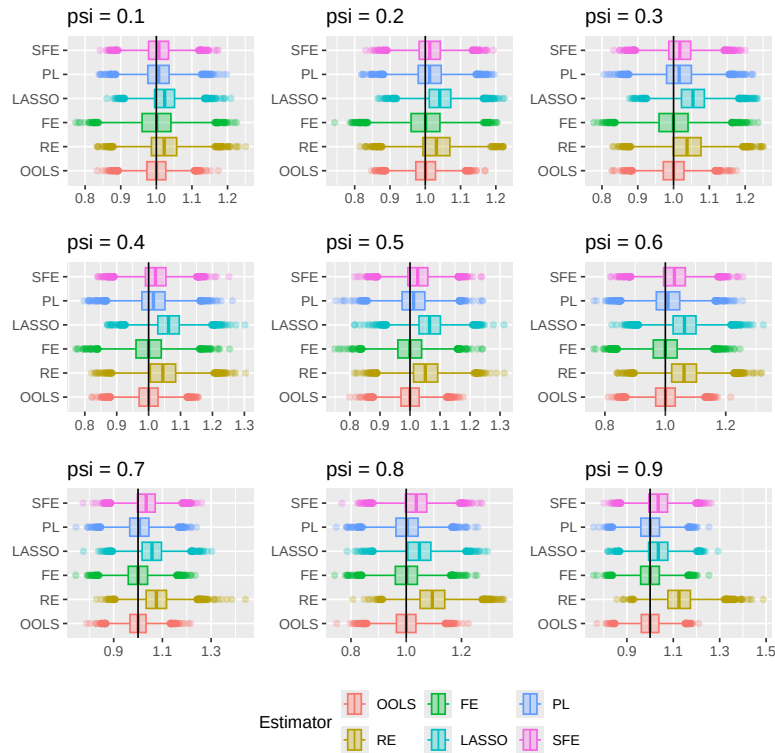


Figure 23: Design B ($\rho = 0.6$)

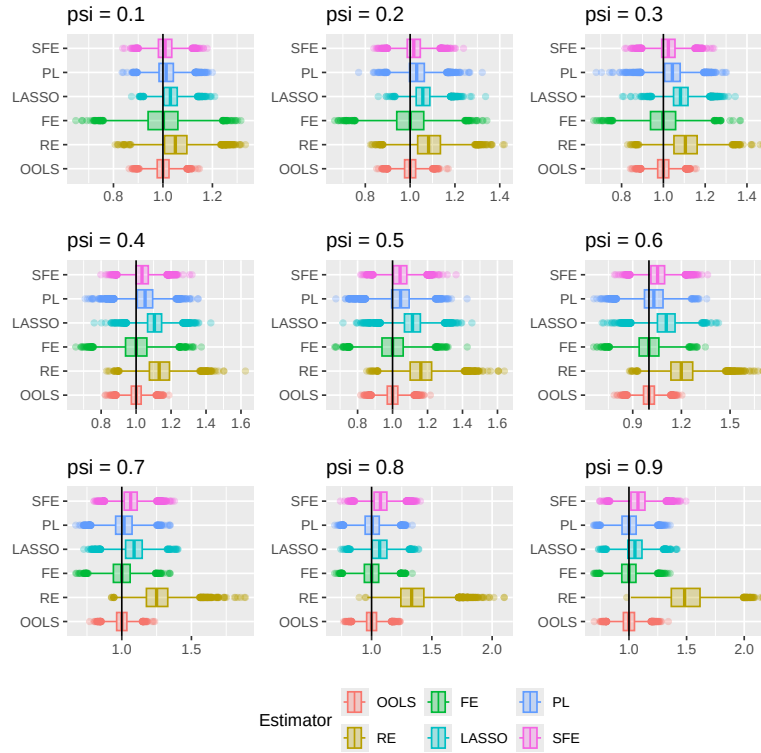


Figure 24: Design B ($\rho = 0.8$)

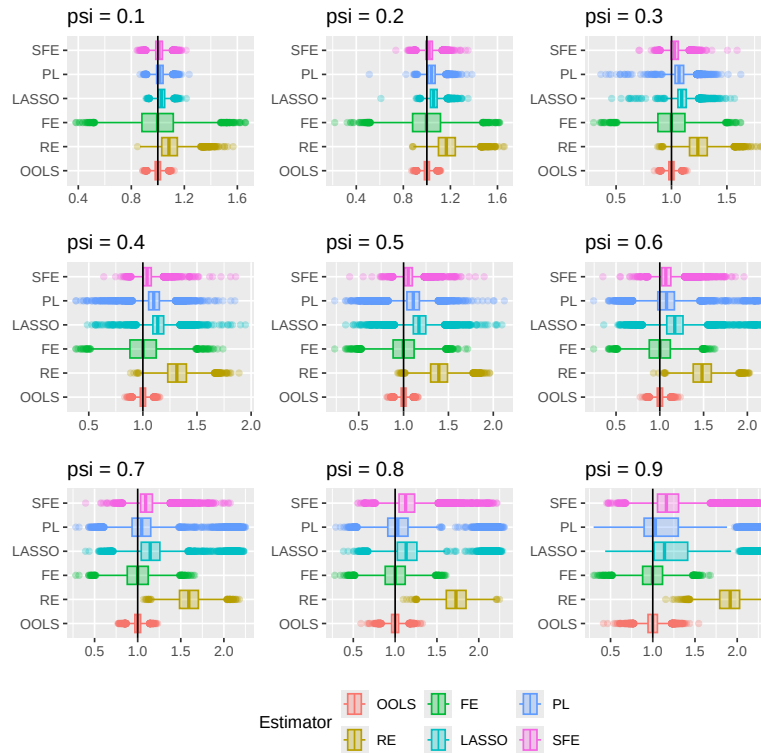


Figure 25: Design C ($\rho = 0$)

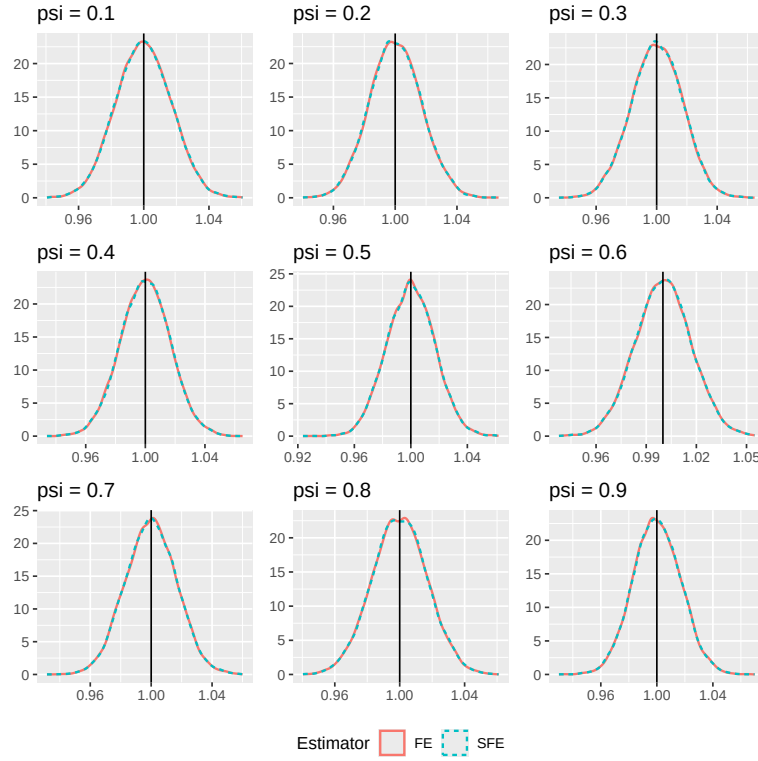


Figure 26: Design C ($\rho = 0.2$)

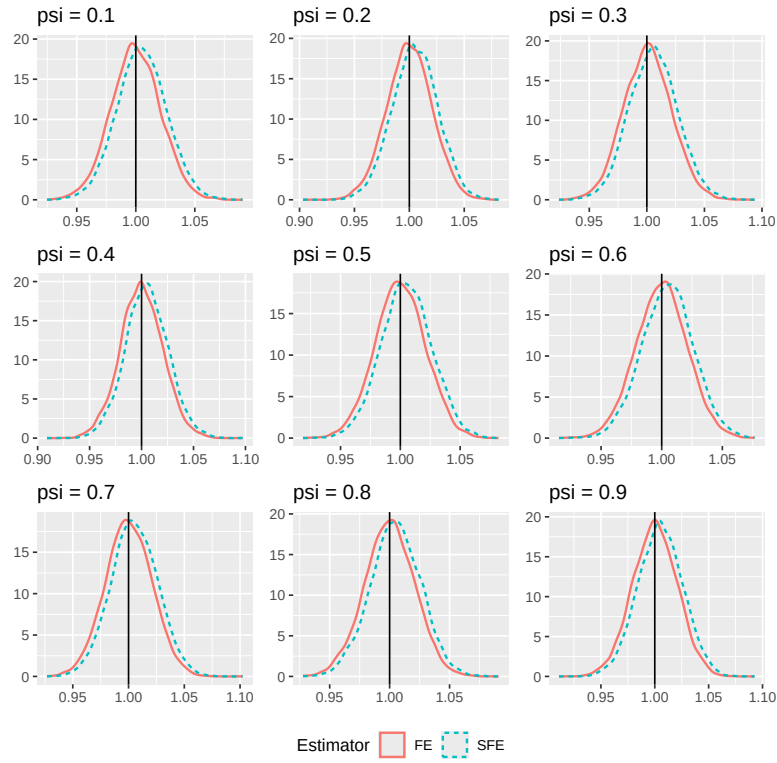


Figure 27: Design C ($\rho = 0.4$)

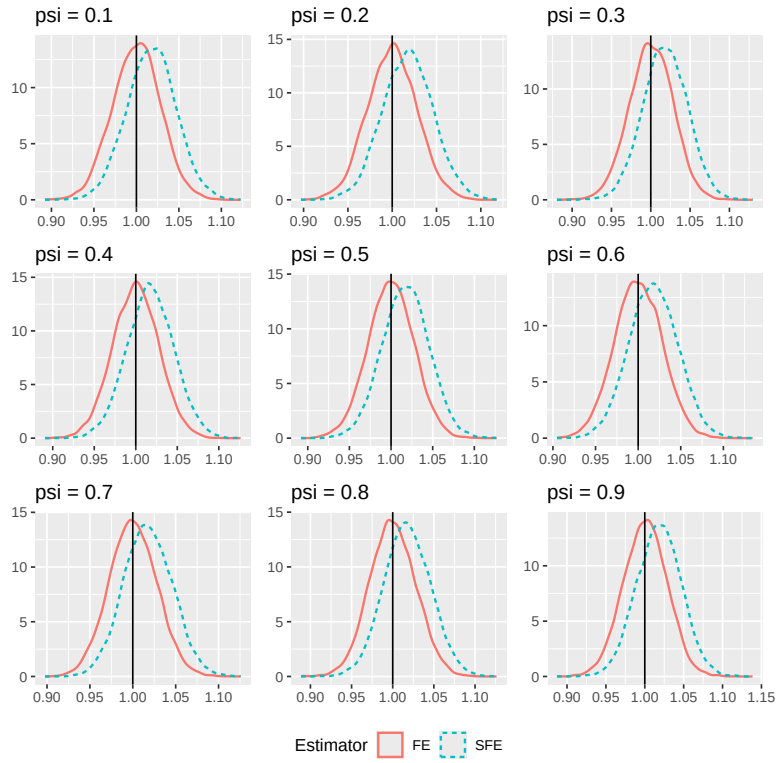


Figure 28: Design C ($\rho = 0.6$)

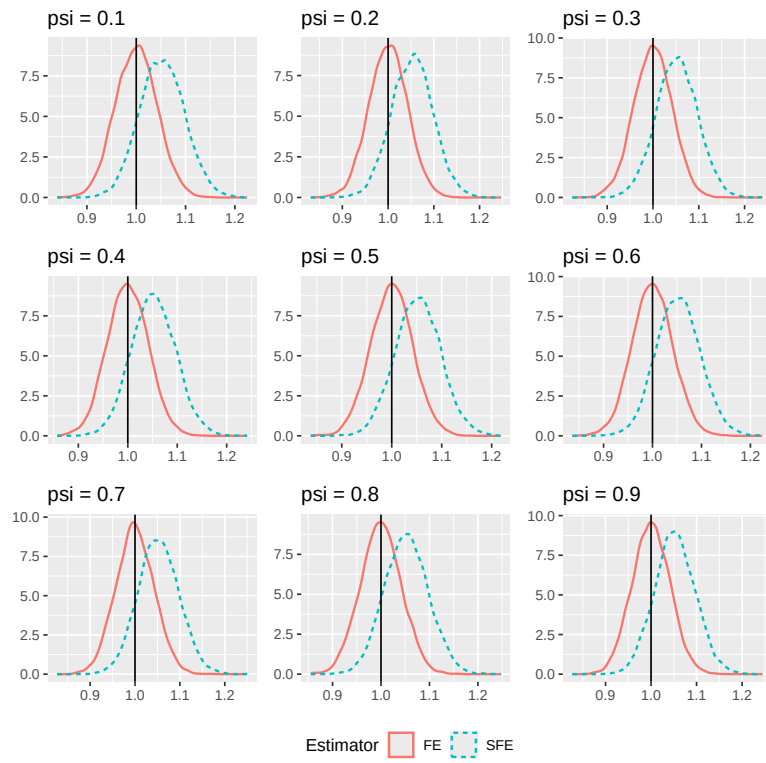


Figure 29: Design C ($\rho = 0.8$)

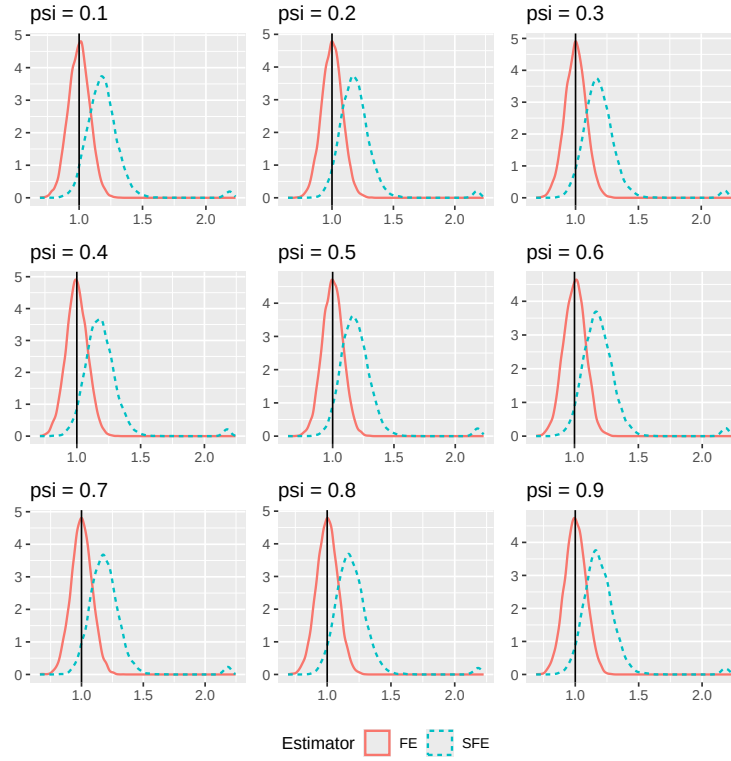


Figure 30: Design D ($\rho = 0$)

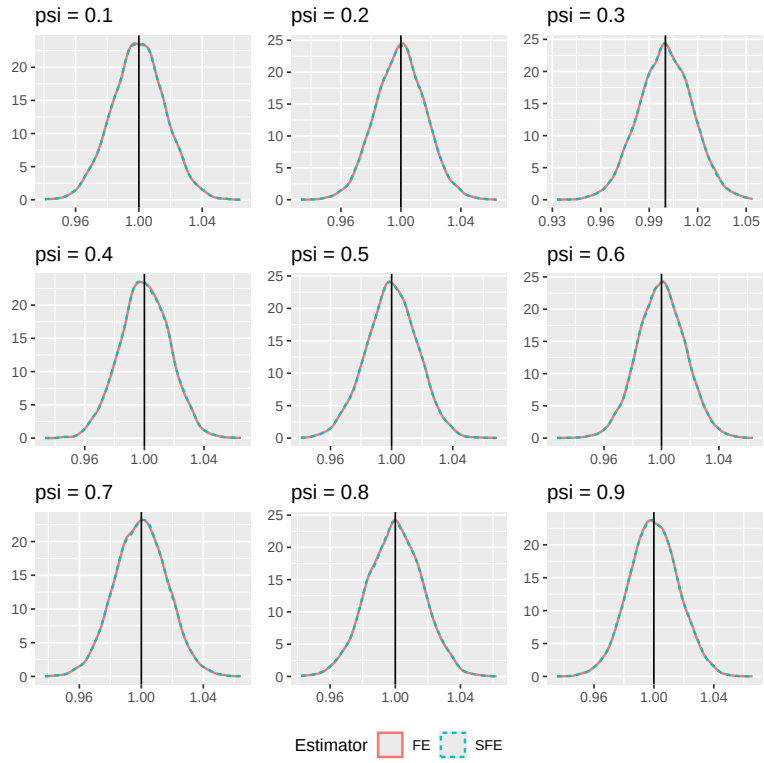


Figure 31: Design D ($\rho = 0.2$)

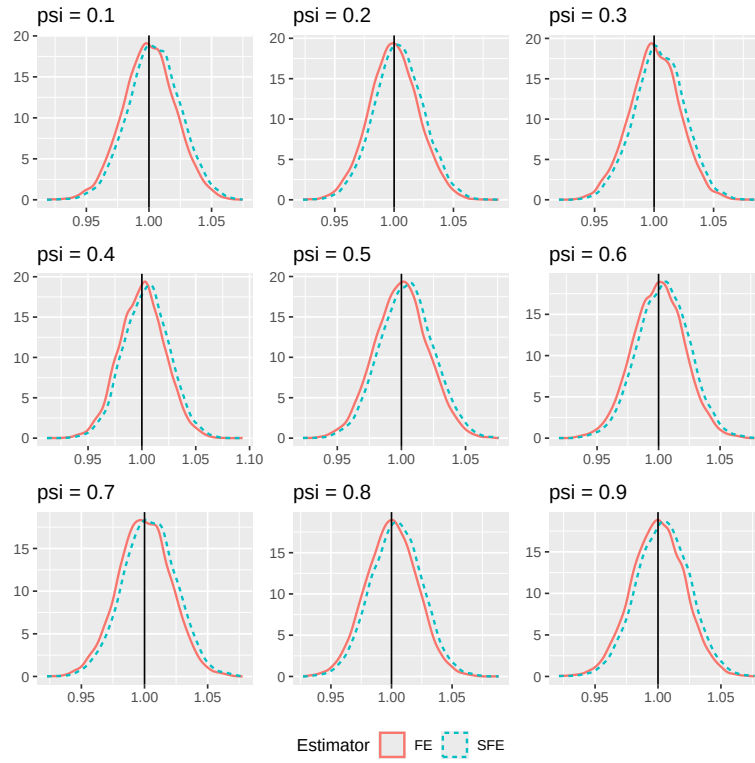


Figure 32: Design D ($\rho = 0.4$)

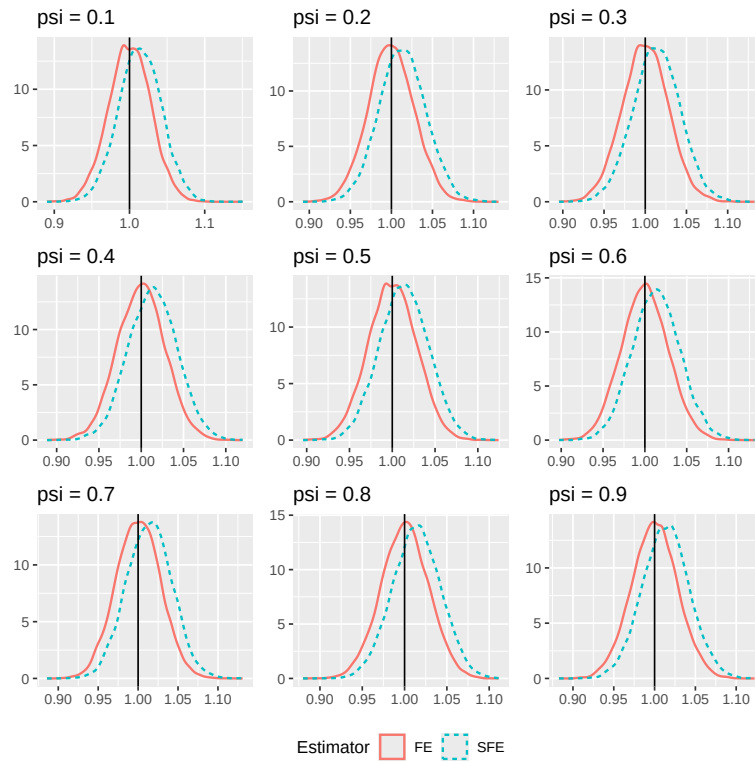


Figure 33: Design D ($\rho = 0.6$)

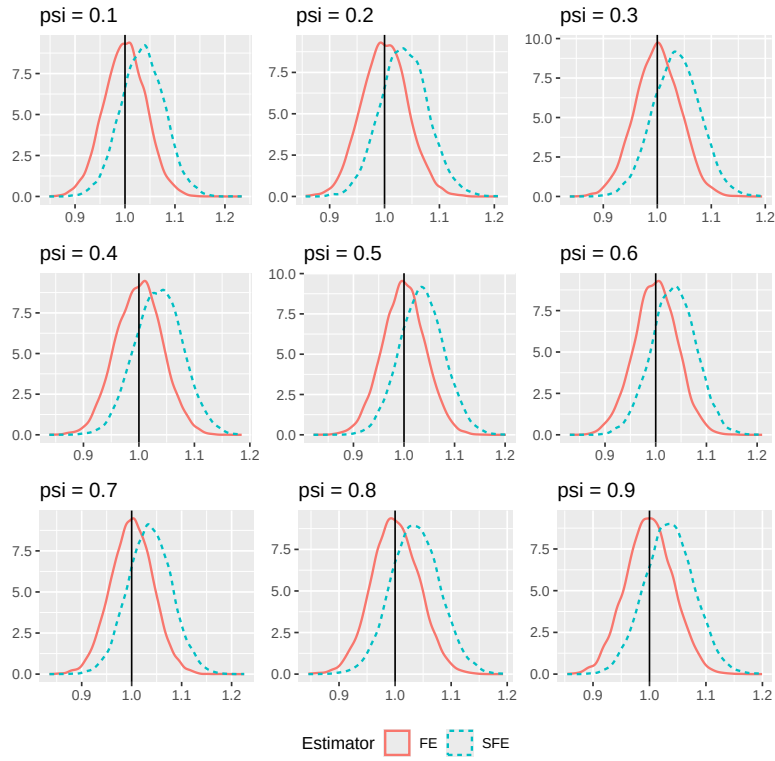
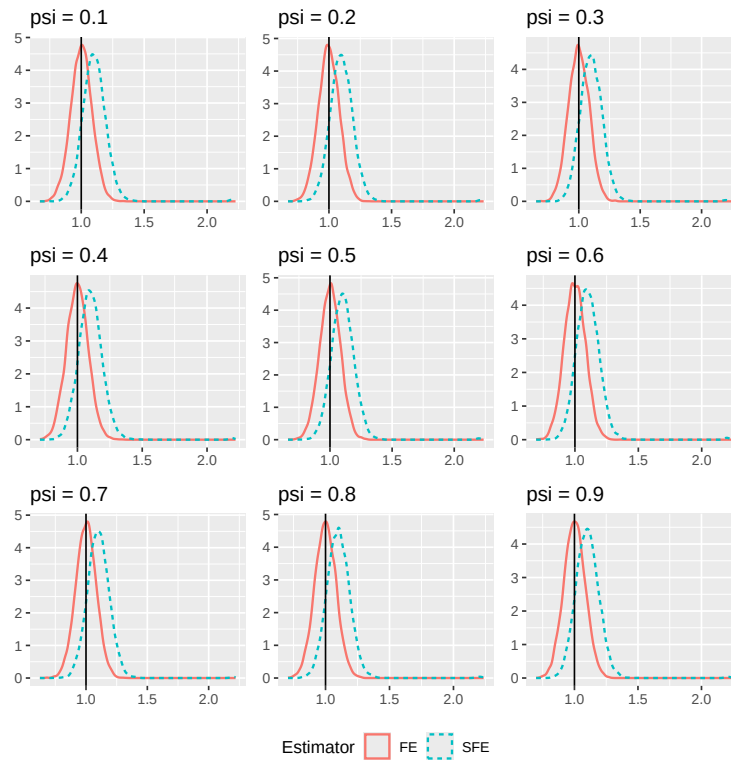


Figure 34: Design D ($\rho = 0.8$)



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